EDA_SS

```
library(haven)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(janitor)
```

```
Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```
#
parent_dir <- dirname(getwd())
# data_folder <- file.path(parent_dir, "data")
setwd("~/ghp525-econometrics-for-health-policy-group-project")

dat25 <- read_xpt("data/LLCP2025.XPT ")
```

```r
dat24 <- read_xpt("data/LLCP2024.XPT ")
# dat23 is not used in the current five-wave analytic sample.
# dat23 <- read_xpt("data/LLCP2023.XPT ")
dat22 <- read_xpt("data/LLCP2022.XPT ")
# dat21 is not used in the current five-wave analytic sample.
# dat21 <- read_xpt("data/LLCP2021.XPT ")
# dat20 is not used in the current five-wave analytic sample.
# dat20 <- read_xpt("data/LLCP2020.XPT ")
dat19 <- read_xpt("data/LLCP2019.XPT ")
# dat18 is not used in the current five-wave analytic sample.
# dat18 <- read_xpt("data/LLCP2018.XPT ")
dat17 <- read_xpt("data/LLCP2017.XPT ")
```

Exploratory data analysis

variables asked only in 2025, 2024, 2022, 2019, and 2017

`NOBCUSE9` = What was main reason for not doing anything to keep you from getting pregnant? `PFPPRVN4` = The last time you had sexual intercourse, did you or your partner do anything to keep you from getting pregnant? () `FAMILY PLANNING` = The last time you had sexual intercourse, what did you or your partner do to keep you from getting pregnant? (lots of response categories!!!)

variables asked in all surveys

`SEXVAR` = 2 if female

`MARITAL`= marital status (1=married, 2=divorced, 3=widowed, 4=separated, 5=never married,6=A member of an unmarried couple, 9 =refused)

`HOME`= homeowner status (1=own, 2=rent, 3=other arrangement, 7=don't know, 9=refused)

`PREGNANT` = current pregnancy status (1=yes, 2=no, 7=don't know, 9=refused)

`MENTHLTH` = Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good? (1-30, 88=none,77=don't know, 99=refused)

`EMPLOY` = employment status (1 = employed for wages, 2 = self-employed, 3 = out of work for 1 year or more, 4 = out of work for less than 1 year, 5 = homemaker, 6 = student, 7 = retired, 8 = unable to work, 9 = refused)

`INCOME` = income categories (1=less than 15k, 2=15k to 25k, 3= 25k to 35k, 4= 35k to 50k, 5=50k to 100k, 6=100k to 200k, 7=200k or more, 9=don't know)

`EDUCA` = What is the highest grade or year of school you completed? (1=no high school, 2=high school, 3=some college,4=college,9=don't know)

`INSURANCE` = Respondents aged 18-64 who have any form of health insurance (1=yes, 2=no, 9=don't know)

`RACE` = Race/ethnicity categories (1=nh white, 2=nh black, 3=nh native american, 4=nh asian, 5=nh native hawaiian, 7=nh multiracial,9=don't know)

subset each survey year to include relevant variables, subsetting the sample to only women

Recode the states in each year since states that didn't collect enough data for a given year weren't included in the dataset

```
sub_dat25 <- dat25 |> select(c("_STATE", "_AGEG5YR", "SEXVAR", "PREGNANT",
 ↪  "MENTHLTH", "EMPLOY1","_INCOMG1", "EDUCA", "_HCVU654", "_RACE",
 ↪  "RENTHOM1", "MARITAL", "HADSEX", "PFPPRVN4", "TYPCNT10", "NOBCUSE9")) |>
 ↪  filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG1',
 ↪  INSURANCE='_HCVU654', EMPLOY='EMPLOY1', FAMILY_PLANNING="TYPCNT10",
 ↪  PREVENT_PREGNANCY="PFPPRVN4", NO_PREVENT_REASON="NOBCUSE9",
 ↪  "HOME"=RENTHOM1, RACE="_RACE")

state_labels_2025 <- c(
  `1`  = "Alabama",
  `2`  = "Alaska",
  `4`  = "Arizona",
  `5`  = "Arkansas",
  `8`  = "Colorado",
  `9`  = "Connecticut",
  `10` = "Delaware",
  `11` = "District of Columbia",
  `12` = "Florida",
  `13` = "Georgia",
  `15` = "Hawaii",
  `16` = "Idaho",
  `17` = "Illinois",
  `18` = "Indiana",
  `19` = "Iowa",
  `20` = "Kansas",
```

```
  `21` = "Kentucky",
  `22` = "Louisiana",
  `23` = "Maine",
  `24` = "Maryland",
  `25` = "Massachusetts",
  `26` = "Michigan",
  `27` = "Minnesota",
  `29` = "Missouri",
  `30` = "Montana",
  `31` = "Nebraska",
  `33` = "New Hampshire",
  `34` = "New Jersey",
  `35` = "New Mexico",
  `36` = "New York",
  `37` = "North Carolina",
  `38` = "North Dakota",
  `39` = "Ohio",
  `40` = "Oklahoma",
  `41` = "Oregon",
  `42` = "Pennsylvania",
  `44` = "Rhode Island",
  `45` = "South Carolina",
  `46` = "South Dakota",
  `47` = "Tennessee",
  `48` = "Texas",
  `49` = "Utah",
  `50` = "Vermont",
  `51` = "Virginia",
  `53` = "Washington",
  `54` = "West Virginia",
  `55` = "Wisconsin",
  `56` = "Wyoming",
  `66` = "Guam",
  `72` = "Puerto Rico"
)

sub_dat25 <- sub_dat25 |>
  dplyr::mutate(
    `_STATE` = as.character(unname(
      state_labels_2025[as.character(`_STATE`)]
    )
  )) |>
```

```
    mutate(EDUCA = case_when(
        EDUCA %in% 1:3 ~ 1, # no high school
        EDUCA == 4 ~ 2, # high school
        EDUCA == 5 ~ 3 ,#Some college
        EDUCA == 6 ~ 4 ,#College graduate
        EDUCA == 9 ~ 9  #Don't know
  ))
```

```
sub_dat24 <- dat24 |> select(c("_STATE","_AGEG5YR","SEXVAR","PREGNANT",
 ↪  "MENTHLTH", "EMPLOY1","_INCOMG1", "EDUCA", "_HCVU654","_RACE",
 ↪  "RENTHOM1", "MARITAL", "TYPCNTR9")) |> filter(SEXVAR==2) |>
 ↪  select(-SEXVAR) |> rename(INCOME='_INCOMG1', INSURANCE='_HCVU654',
 ↪  EMPLOY='EMPLOY1', FAMILY_PLANNING="TYPCNTR9", "HOME"=RENTHOM1,
 ↪  RACE="_RACE")

state_labels_2024 <- c(
  `1`  = "Alabama",
  `2`  = "Alaska",
  `4`  = "Arizona",
  `5`  = "Arkansas",
  `6`  = "California",
  `8`  = "Colorado",
  `9`  = "Connecticut",
  `10` = "Delaware",
  `11` = "District of Columbia",
  `12` = "Florida",
  `13` = "Georgia",
  `15` = "Hawaii",
  `16` = "Idaho",
  `17` = "Illinois",
  `18` = "Indiana",
  `19` = "Iowa",
  `20` = "Kansas",
  `21` = "Kentucky",
  `22` = "Louisiana",
  `23` = "Maine",
  `24` = "Maryland",
  `25` = "Massachusetts",
  `26` = "Michigan",
  `27` = "Minnesota",
  `28` = "Mississippi",
```

```
  `29` = "Missouri",
  `30` = "Montana",
  `31` = "Nebraska",
  `32` = "Nevada",
  `33` = "New Hampshire",
  `34` = "New Jersey",
  `35` = "New Mexico",
  `36` = "New York",
  `37` = "North Carolina",
  `38` = "North Dakota",
  `39` = "Ohio",
  `40` = "Oklahoma",
  `41` = "Oregon",
  `42` = "Pennsylvania",
  `44` = "Rhode Island",
  `45` = "South Carolina",
  `46` = "South Dakota",
  `48` = "Texas",
  `49` = "Utah",
  `50` = "Vermont",
  `51` = "Virginia",
  `53` = "Washington",
  `54` = "West Virginia",
  `55` = "Wisconsin",
  `56` = "Wyoming",
  `66` = "Guam",
  `72` = "Puerto Rico",
  `78` = "Virgin Islands"
)
sub_dat24 <- sub_dat24 |>
  dplyr::mutate(
    `_STATE` = as.character(unname(
      state_labels_2024[as.character(`_STATE`)]
    )
  )) |> mutate(EDUCA = case_when(
        EDUCA %in% 1:3 ~ 1, # no high school
        EDUCA == 4 ~ 2, # high school
        EDUCA == 5 ~ 3 ,#Some college
        EDUCA == 6 ~ 4 ,#College graduate
        EDUCA == 9 ~ 9  #Don't know
  ))
```

```r
# Roe v. Wade was overturned on June 24, 2022
sub_dat22 <- dat22 |> select(c("_STATE","_AGEG5YR","SEXVAR","IMONTH","IDAY",
 ↪  "IYEAR","PREGNANT", "MENTHLTH", "EMPLOY1","_INCOMG1", "EDUCA",
 ↪  "_HCVU652", "_RACE1", "TYPCNTR9", "RENTHOM1", "MARITAL")) |>
 ↪  filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG1',
 ↪  INSURANCE='_HCVU652', EMPLOY='EMPLOY1', FAMILY_PLANNING="TYPCNTR9",
 ↪  HOME="RENTHOM1", RACE="_RACE1")
state_labels_2022 <- c(
  `1`  = "Alabama",
  `2`  = "Alaska",
  `4`  = "Arizona",
  `5`  = "Arkansas",
  `6`  = "California",
  `8`  = "Colorado",
  `9`  = "Connecticut",
  `10` = "Delaware",
  `11` = "District of Columbia",
  `12` = "Florida",
  `13` = "Georgia",
  `15` = "Hawaii",
  `16` = "Idaho",
  `17` = "Illinois",
  `18` = "Indiana",
  `19` = "Iowa",
  `20` = "Kansas",
  `21` = "Kentucky",
  `22` = "Louisiana",
  `23` = "Maine",
  `24` = "Maryland",
  `25` = "Massachusetts",
  `26` = "Michigan",
  `27` = "Minnesota",
  `28` = "Mississippi",
  `29` = "Missouri",
  `30` = "Montana",
  `31` = "Nebraska",
  `32` = "Nevada",
  `33` = "New Hampshire",
  `34` = "New Jersey",
  `35` = "New Mexico",
  `36` = "New York",
  `37` = "North Carolina",
```

```
  `38` = "North Dakota",
  `39` = "Ohio",
  `40` = "Oklahoma",
  `41` = "Oregon",
  `42` = "Pennsylvania",
  `44` = "Rhode Island",
  `45` = "South Carolina",
  `46` = "South Dakota",
  `47` = "Tennessee",
  `48` = "Texas",
  `49` = "Utah",
  `50` = "Vermont",
  `51` = "Virginia",
  `53` = "Washington",
  `54` = "West Virginia",
  `55` = "Wisconsin",
  `56` = "Wyoming",
  `66` = "Guam",
  `72` = "Puerto Rico",
  `78` = "Virgin Islands"
)
sub_dat22 <- sub_dat22 |>
  dplyr::mutate(
    `_STATE` = as.character(unname(
      state_labels_2022[as.character(`_STATE`)]
    )
  ))|> mutate(EDUCA = case_when(
        EDUCA %in% 1:3 ~ 1, # no high school
        EDUCA == 4 ~ 2, # high school
        EDUCA == 5 ~ 3 ,#Some college
        EDUCA == 6 ~ 4 ,#College graduate
        EDUCA == 9 ~ 9  #Don't know
  ))
```

```
sub_dat19 <- dat19 |> select(c("_STATE","_AGEG5YR","SEXVAR","PREGNANT","MENT
↪  HLTH","EMPLOY1","_INCOMG", "EDUCA", "_HCVU651", "_RACE", "RENTHOM1",
↪  "MARITAL", "TYPCNTR8")) |> filter(SEXVAR==2) |> select(-SEXVAR) |>
↪  rename(INCOME='_INCOMG', INSURANCE='_HCVU651', EMPLOY='EMPLOY1',
↪  FAMILY_PLANNING="TYPCNTR8", "HOME"=RENTHOM1, RACE="_RACE")
state_labels_2019 <- c(
  `1`  = "Alabama",
```

```
  `2`  = "Alaska",
  `4`  = "Arizona",
  `5`  = "Arkansas",
  `6`  = "California",
  `8`  = "Colorado",
  `9`  = "Connecticut",
  `10` = "Delaware",
  `11` = "District of Columbia",
  `12` = "Florida",
  `13` = "Georgia",
  `15` = "Hawaii",
  `16` = "Idaho",
  `17` = "Illinois",
  `18` = "Indiana",
  `19` = "Iowa",
  `20` = "Kansas",
  `21` = "Kentucky",
  `22` = "Louisiana",
  `23` = "Maine",
  `24` = "Maryland",
  `25` = "Massachusetts",
  `26` = "Michigan",
  `27` = "Minnesota",
  `28` = "Mississippi",
  `29` = "Missouri",
  `30` = "Montana",
  `31` = "Nebraska",
  `32` = "Nevada",
  `33` = "New Hampshire",
  `35` = "New Mexico",
  `36` = "New York",
  `37` = "North Carolina",
  `38` = "North Dakota",
  `39` = "Ohio",
  `40` = "Oklahoma",
  `41` = "Oregon",
  `42` = "Pennsylvania",
  `44` = "Rhode Island",
  `45` = "South Carolina",
  `46` = "South Dakota",
  `47` = "Tennessee",
  `48` = "Texas",
```

```
  `49` = "Utah",
  `50` = "Vermont",
  `51` = "Virginia",
  `53` = "Washington",
  `54` = "West Virginia",
  `55` = "Wisconsin",
  `56` = "Wyoming",
  `66` = "Guam",
  `72` = "Puerto Rico"
)
sub_dat19 <- sub_dat19 |>
  dplyr::mutate(
    `_STATE` = as.character(unname(
      state_labels_2019[as.character(`_STATE`)]
    )
  )) |>
    mutate(EDUCA = case_when(
        EDUCA %in% 1:3 ~ 1, # no high school
        EDUCA == 4 ~ 2, # high school
        EDUCA == 5 ~ 3 ,#Some college
        EDUCA == 6 ~ 4 ,#College graduate
        EDUCA == 9 ~ 9  #Don't know
  ))
```

```
sub_dat17 <- dat17 |> select(c("_STATE", "_AGEG5YR", "SEX","PREGNANT",
 ↪  "MENTHLTH", "EMPLOY1", "_INCOMG", "EDUCA", "_HCVU651",  "_RACE",
 ↪  "RENTHOM1", "MARITAL", "TYPCNTR7")) |> filter(SEX==2) |> select(-SEX) |>
 ↪  rename(INCOME='_INCOMG', INSURANCE='_HCVU651', EMPLOY='EMPLOY1',
 ↪  FAMILY_PLANNING="TYPCNTR7", "HOME"=RENTHOM1, RACE="_RACE")
state_labels_2017 <- c(
  `1`  = "Alabama",
  `2`  = "Alaska",
  `4`  = "Arizona",
  `5`  = "Arkansas",
  `6`  = "California",
  `8`  = "Colorado",
  `9`  = "Connecticut",
  `10` = "Delaware",
  `11` = "District of Columbia",
  `12` = "Florida",
  `13` = "Georgia",
```

```
  `15` = "Hawaii",
  `16` = "Idaho",
  `17` = "Illinois",
  `18` = "Indiana",
  `19` = "Iowa",
  `20` = "Kansas",
  `21` = "Kentucky",
  `22` = "Louisiana",
  `23` = "Maine",
  `24` = "Maryland",
  `25` = "Massachusetts",
  `26` = "Michigan",
  `27` = "Minnesota",
  `28` = "Mississippi",
  `29` = "Missouri",
  `30` = "Montana",
  `31` = "Nebraska",
  `32` = "Nevada",
  `33` = "New Hampshire",
  `34` = "New Jersey",
  `35` = "New Mexico",
  `36` = "New York",
  `37` = "North Carolina",
  `38` = "North Dakota",
  `39` = "Ohio",
  `40` = "Oklahoma",
  `41` = "Oregon",
  `42` = "Pennsylvania",
  `44` = "Rhode Island",
  `45` = "South Carolina",
  `46` = "South Dakota",
  `47` = "Tennessee",
  `48` = "Texas",
  `49` = "Utah",
  `50` = "Vermont",
  `51` = "Virginia",
  `53` = "Washington",
  `54` = "West Virginia",
  `55` = "Wisconsin",
  `56` = "Wyoming",
  `66` = "Guam",
  `72` = "Puerto Rico"
```

```
)
sub_dat17 <- sub_dat17 |>
  dplyr::mutate(
    `_STATE` = as.character(unname(
      state_labels_2017[as.character(`_STATE`)]
    )
  )) |>
  mutate(EDUCA = case_when(
        EDUCA %in% 1:3 ~ 1, # no high school
        EDUCA == 4 ~ 2, # high school
        EDUCA == 5 ~ 3 ,#Some college
        EDUCA == 6 ~ 4 ,#College graduate
        EDUCA == 9 ~ 9  #Don't know
  ))
```

```
sub_dat23 <- dat23 |> select(c("_STATE","_AGEG5YR","SEXVAR","PREGNANT",
 ↪  "MENTHLTH","EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU653", "_RACE")) |>
 ↪  filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG1',
 ↪  INSURANCE='_HCVU653', EMPLOY='EMPLOY1')


sub_dat21 <- dat21 |> select(c("_STATE","_AGEG5YR","SEXVAR","PREGNANT",
 ↪  "MENTHLTH","EMPLOY1","_INCOMG1", "EDUCA","_HCVU652", "_RACE")) |>
 ↪  filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG1',
 ↪  INSURANCE='_HCVU652', EMPLOY='EMPLOY1')

sub_dat20 <- dat20 |> select(c("_STATE","_AGEG5YR","SEXVAR","PREGNANT",
 ↪  "MENTHLTH","EMPLOY1", "_INCOMG", "EDUCA",  "_HCVU651", "_RACE")) |>
 ↪  filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG',
 ↪  INSURANCE='_HCVU651', EMPLOY='EMPLOY1')


sub_dat18 <- dat18 |> select(c("_STATE", "SEX1","PREGNANT", "MENTHLTH",
 ↪  "EMPLOY1", "_INCOMG", "EDUCA", "_HCVU651",  "_RACE", "RENTHOM1",
 ↪  "MARITAL")) |> filter(SEX1==2) |> select(-SEX1) |>
 ↪  rename(INCOME='_INCOMG', INSURANCE='_HCVU651', EMPLOY='EMPLOY1')
```

merge relevant questionnaire years

```
data_list <- list(
  `2017` = sub_dat17,
  `2019` = sub_dat19,
  `2022` = sub_dat22,
  `2024` = sub_dat24,
  `2025` = sub_dat25
)

vars_to_keep <- c(
  "_STATE",
  "_AGEG5YR",
  "PREGNANT",
  "MENTHLTH",
  "EMPLOY",
  "INCOME",
  "EDUCA",
  "INSURANCE",
  "RACE",
  "MARITAL",
  "HOME",
  "FAMILY_PLANNING",
  "IYEAR",
  "IMONTH",
  "IDAY"
)

analysis_data <- purrr::imap_dfr(
  data_list,
  function(dataset, year) {

    year <- as.integer(year)

    dataset <- dataset |>
      dplyr::select(dplyr::any_of(vars_to_keep)) |>
      dplyr::mutate(
        YEAR = year
      )

    if (year == 2022) {
```

```
      dataset <- dataset |>
        dplyr::mutate(
          IYEAR = as.integer(as.character(IYEAR)),
          IMONTH = as.integer(as.character(IMONTH)),
          IDAY = as.integer(as.character(IDAY)),

          date = as.Date(
            sprintf("%04d-%02d-%02d", IYEAR, IMONTH, IDAY)
          ),

          post_ban = dplyr::case_when(
            date < as.Date("2022-06-24") ~ 0L,
            date >= as.Date("2022-06-24") ~ 1L,
            TRUE ~ NA_integer_
          )
        )
    } else {
      dataset <- dataset |>
        dplyr::mutate(
          date = as.Date(NA),

          POST_BAN = dplyr::case_when(
            YEAR <= 2021 ~ 0L,
            YEAR >= 2023 ~ 1L,
            TRUE ~ NA_integer_
          )
        )
    }

    dataset
  }
) |>
  dplyr::rename(STATE = `_STATE`) |> select(-c(IYEAR, IMONTH, IDAY, date))


# check that post ban was correctly created
analysis_data |> filter(YEAR == 2025) |> janitor::tabyl(post_ban)
```

```
 post_ban      n percent valid_percent
       NA 187696       1            NA
```

```r
analysis_data |> filter(YEAR == 2024) |> janitor::tabyl(post_ban)
```

```
 post_ban      n percent valid_percent
       NA 240183       1            NA
```

```r
analysis_data |> filter(YEAR == 2022) |> janitor::tabyl(post_ban)
```

```
 post_ban      n      percent valid_percent
        0  96964 4.110508e-01      0.411056
        1 138926 5.889365e-01      0.588944
       NA      3 1.271763e-05            NA
```

```r
analysis_data |> filter(YEAR == 2019) |> janitor::tabyl(post_ban)
```

```
 post_ban      n percent valid_percent
       NA 228419       1            NA
```

```r
analysis_data |> filter(YEAR == 2017) |> janitor::tabyl(post_ban)
```

```
 post_ban      n percent valid_percent
       NA 251007       1            NA
```

create U.S. census regions

```r
library(tibble)

census_region_lookup <- tribble(
  ~STATE, ~REGION, ~DIVISION,

  # Northeast
  "Connecticut",          "Northeast", "New England",
  "Maine",                "Northeast", "New England",
  "Massachusetts",        "Northeast", "New England",
  "New Hampshire",        "Northeast", "New England",
  "Rhode Island",         "Northeast", "New England",
  "Vermont",              "Northeast", "New England",
```

```
"New Jersey",           "Northeast", "Middle Atlantic",
"New York",             "Northeast", "Middle Atlantic",
"Pennsylvania",         "Northeast", "Middle Atlantic",

# Midwest
"Illinois",             "Midwest", "East North Central",
"Indiana",              "Midwest", "East North Central",
"Michigan",             "Midwest", "East North Central",
"Ohio",                 "Midwest", "East North Central",
"Wisconsin",            "Midwest", "East North Central",
"Iowa",                 "Midwest", "West North Central",
"Kansas",               "Midwest", "West North Central",
"Minnesota",            "Midwest", "West North Central",
"Missouri",             "Midwest", "West North Central",
"Nebraska",             "Midwest", "West North Central",
"North Dakota",         "Midwest", "West North Central",
"South Dakota",         "Midwest", "West North Central",

# South
"Delaware",             "South", "South Atlantic",
"District of Columbia", "South", "South Atlantic",
"Florida",              "South", "South Atlantic",
"Georgia",              "South", "South Atlantic",
"Maryland",             "South", "South Atlantic",
"North Carolina",       "South", "South Atlantic",
"South Carolina",       "South", "South Atlantic",
"Virginia",             "South", "South Atlantic",
"West Virginia",        "South", "South Atlantic",
"Alabama",              "South", "East South Central",
"Kentucky",             "South", "East South Central",
"Mississippi",          "South", "East South Central",
"Tennessee",            "South", "East South Central",
"Arkansas",             "South", "West South Central",
"Louisiana",            "South", "West South Central",
"Oklahoma",             "South", "West South Central",
"Texas",                "South", "West South Central",

# West
"Arizona",              "West", "Mountain",
"Colorado",             "West", "Mountain",
"Idaho",                "West", "Mountain",
"Montana",              "West", "Mountain",
```

```
  "Nevada",                "West", "Mountain",
  "New Mexico",            "West", "Mountain",
  "Utah",                  "West", "Mountain",
  "Wyoming",               "West", "Mountain",
  "Alaska",                "West", "Pacific",
  "California",            "West", "Pacific",
  "Hawaii",                "West", "Pacific",
  "Oregon",                "West", "Pacific",
  "Washington",            "West", "Pacific",

  # Territories: not assigned to a Census region/division
  "Guam",                  NA_character_, NA_character_,
  "Puerto Rico",           NA_character_, NA_character_,
  "Virgin Islands",        NA_character_, NA_character_
)

# remove Guam, Puerto Rico, and Virgin Islands from analytic sample
analysis_data_v2 <- analysis_data |>
  dplyr::left_join(
    census_region_lookup,
    by = "STATE"
  ) |> filter(!is.na(REGION)) |> rename(AGEG5YR = "_AGEG5YR")
```

```
variable_labels <- list(

  AGEG5YR = c(
    `1` = "18 to 24 years", `2` = "25 to 29 years",
    `3` = "30 to 34 years", `4` = "35 to 39 years",
    `5` = "40 to 44 years", `6` = "45 to 49 years",
    `7` = "50 to 54 years", `8` = "55 to 59 years",
    `9` = "60 to 64 years", `10` = "65 to 69 years",
    `11` = "70 to 74 years", `12` = "75 to 79 years",
    `13` = "80 years or older", `14` = "Don't know/refused/missing"
  ),

  MARITAL = c(
    `1` = "Married",
    `2` = "Divorced",
    `3` = "Widowed",
    `4` = "Separated",
    `5` = "Never married",
```

```
  `6` = "Member of an unmarried couple",
  `9` = "Refused"
),

HOME = c(
  `1` = "Own",
  `2` = "Rent",
  `3` = "Other arrangement",
  `7` = "Don't know",
  `9` = "Refused"
),

PREGNANT = c(
  `1` = "Yes",
  `2` = "No",
  `7` = "Don't know",
  `9` = "Refused"
),

MENTHLTH = c(
  setNames(
    paste(1:30, ifelse(1:30 == 1, "day", "days")),
    as.character(1:30)
  ),
  `77` = "Don't know",
  `88` = "None",
  `99` = "Refused"
),

EMPLOY = c(
  `1` = "Employed for wages",
  `2` = "Self-employed",
  `3` = "Out of work for 1 year or more",
  `4` = "Out of work for less than 1 year",
  `5` = "Homemaker",
  `6` = "Student",
  `7` = "Retired",
  `8` = "Unable to work",
  `9` = "Refused"
),

INCOME = c(
```

```
    `1` = "Less than $15,000",
    `2` = "$15,000 to $24,999",
    `3` = "$25,000 to $34,999",
    `4` = "$35,000 to $49,999",
    `5` = "$50,000 to $99,999",
    `6` = "$100,000 to $199,999",
    `7` = "$200,000 or more",
    `9` = "Don't know"
  ),

  EDUCA = c(
    `1` = "No high school",
    `2` = "High school",
    `3` = "Some college",
    `4` = "College graduate",
    `9` = "Don't know"
  ),

  INSURANCE = c(
    `1` = "Yes",
    `2` = "No",
    `9` = "Don't know"
  ),

  RACE = c(
    `1` = "Non-Hispanic White",
    `2` = "Non-Hispanic Black",
    `3` = "Non-Hispanic Native American",
    `4` = "Non-Hispanic Asian",
    `5` = "Non-Hispanic Native Hawaiian",
    `6` = "Other race, non-Hispanic",
    `7` = "Non-Hispanic Multiracial",
    `8` = "Hispanic",
    `9` = "Don't know"
  )
)

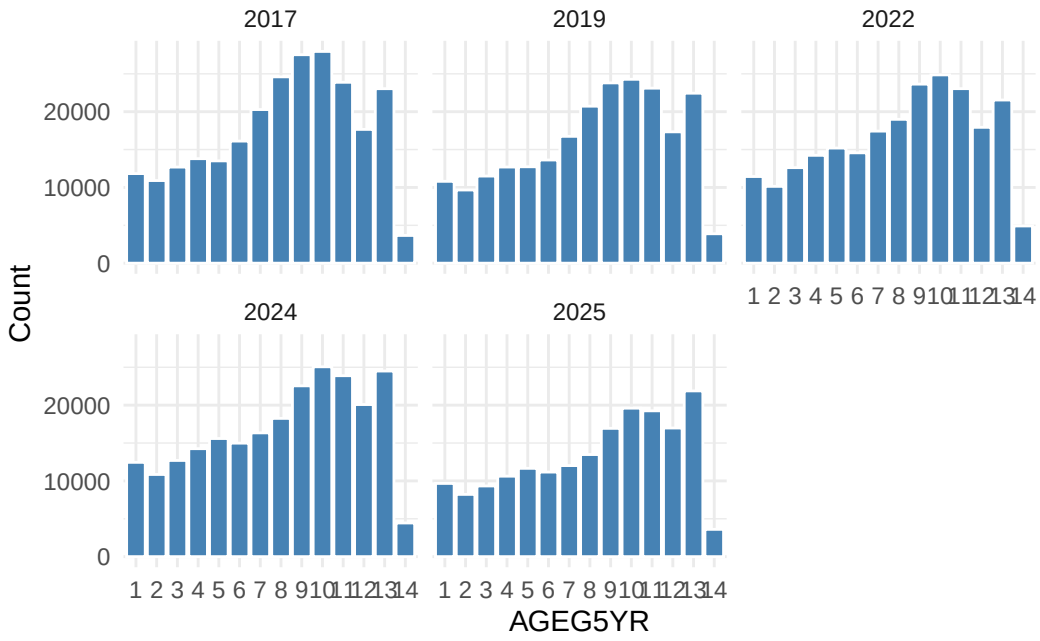
variables_to_summarize <- c(
  "AGEG5YR",
  "MARITAL",
  "HOME",
  "PREGNANT",
```

```
  "MENTHLTH",
  "EMPLOY",
  "INCOME",
  "EDUCA",
  "INSURANCE",
  "RACE"
)
```

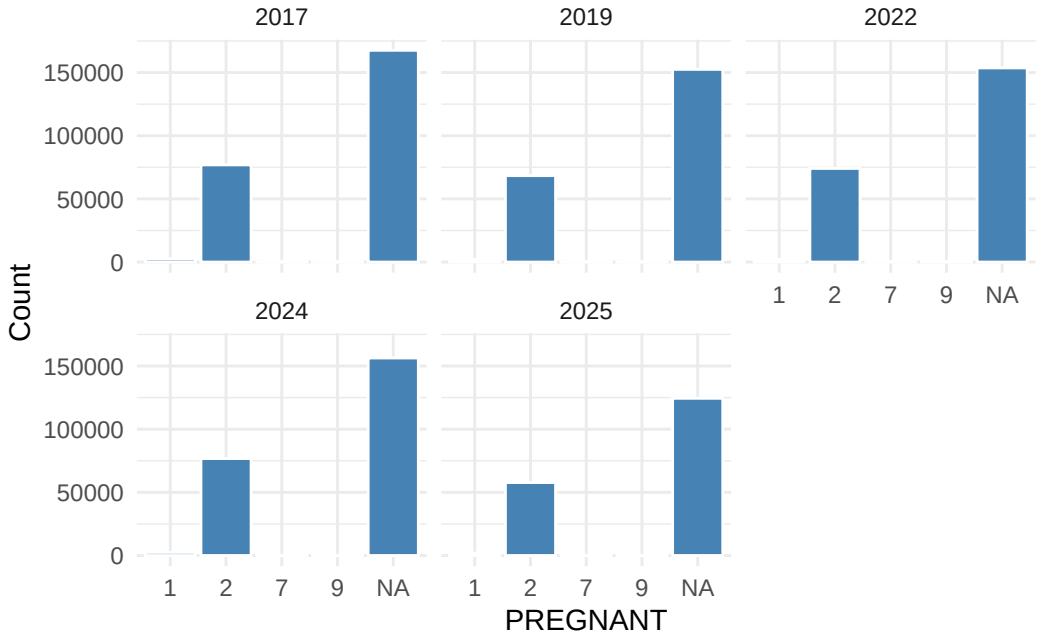
U.S. Population Characteristics Over Time

```
# categorical variables
sapply(names(analysis_data_v2)[!names(analysis_data_v2) %in%
 ↪  c("STATE","YEAR","POST_BAN", "MENTHLTH")], function(var){
  ggplot(
  analysis_data_v2,
  aes(x = factor(.data[[var]]))
) +
  geom_bar(
    fill = "steelblue",
    color = "white"
  ) +
  facet_wrap(~ YEAR) +
  theme_minimal() +
  labs(
    x = var,
    y = "Count"
  )
})
```

```
$AGEG5YR
```



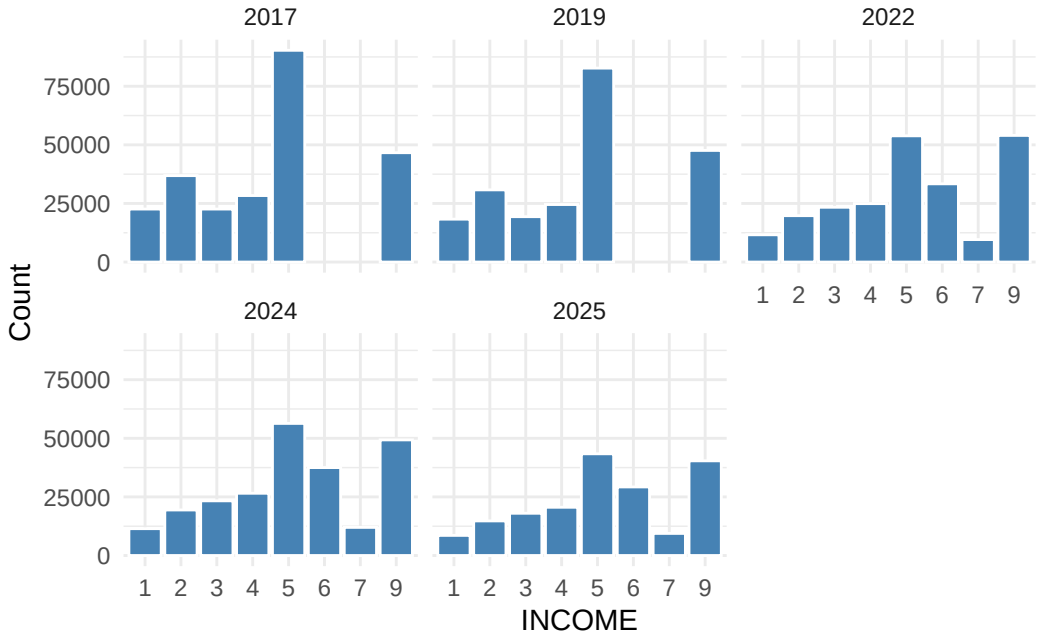
$PREGNANT



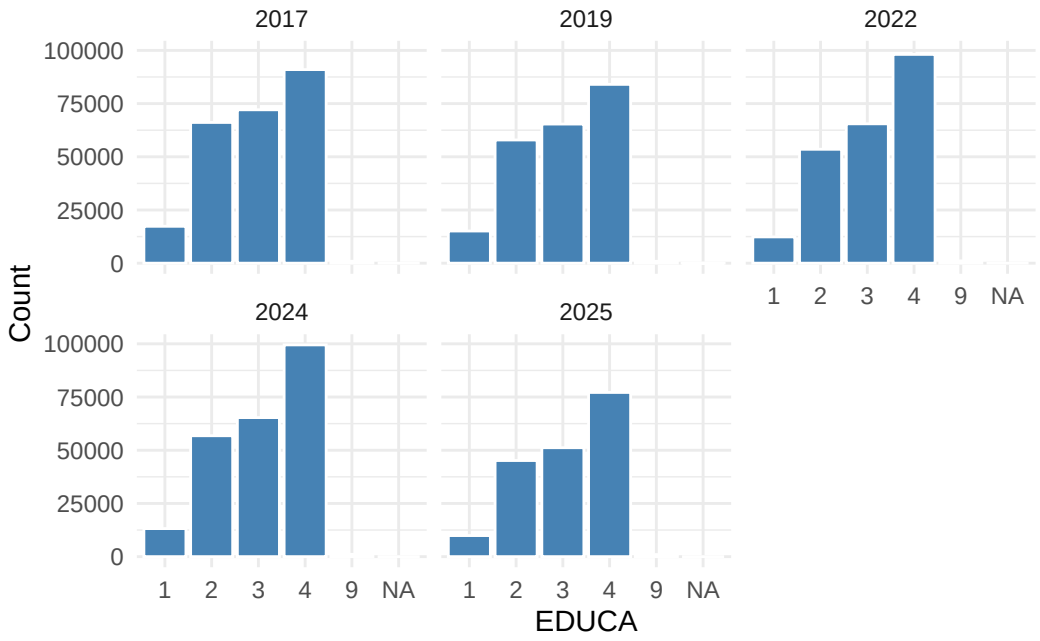
$EMPLOY



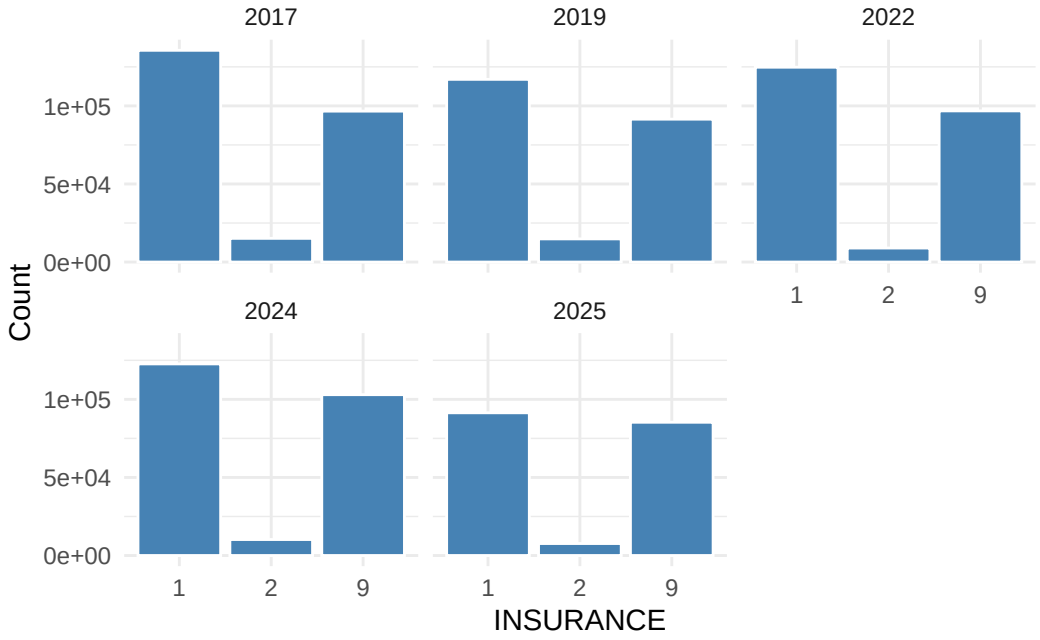
$INCOME



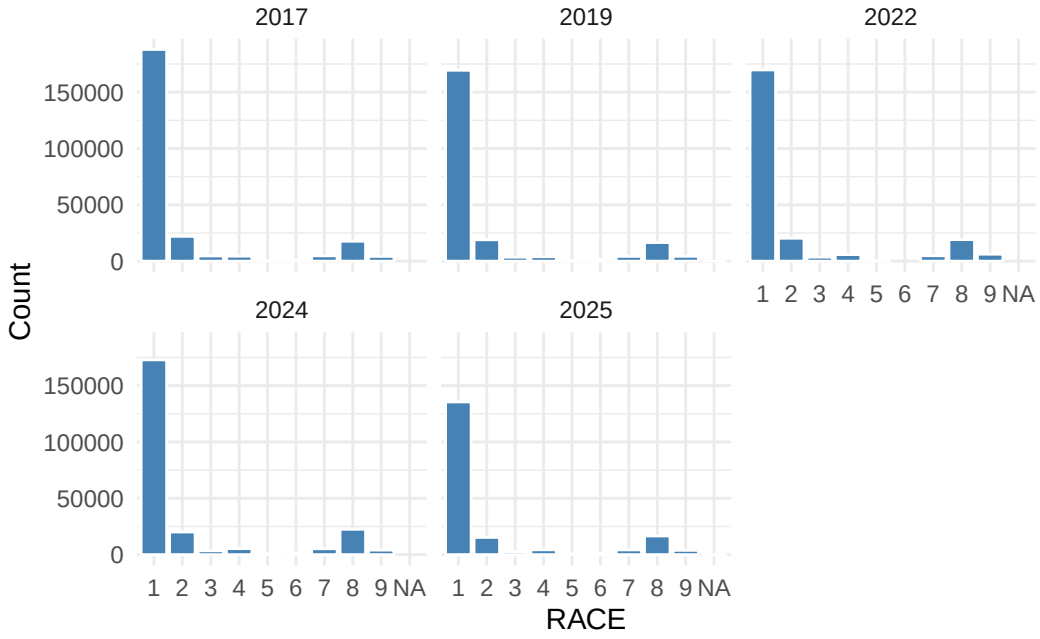
$EDUCA



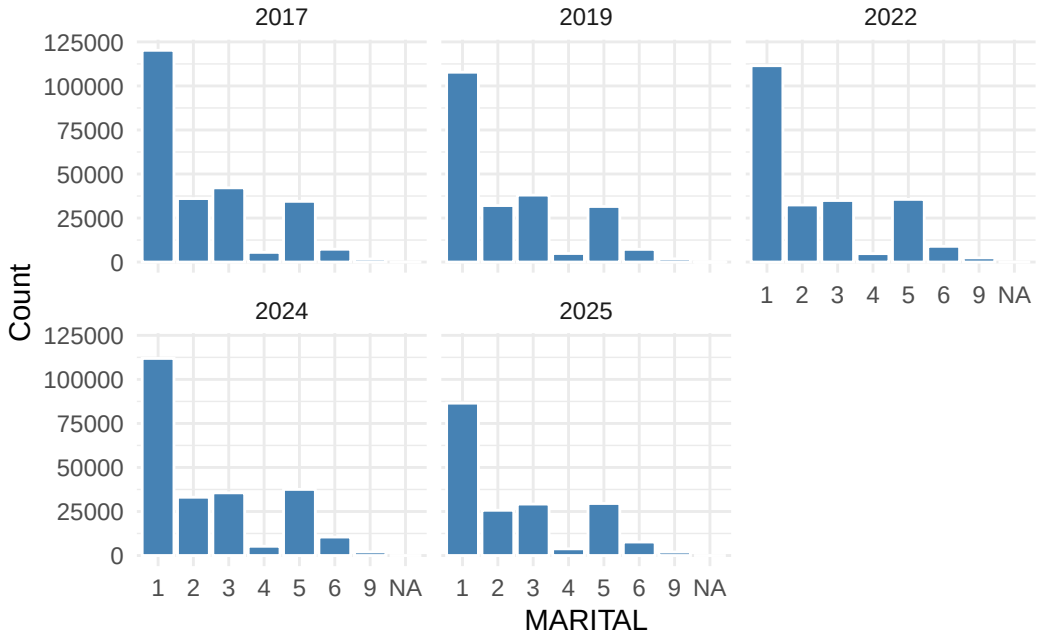
$INSURANCE



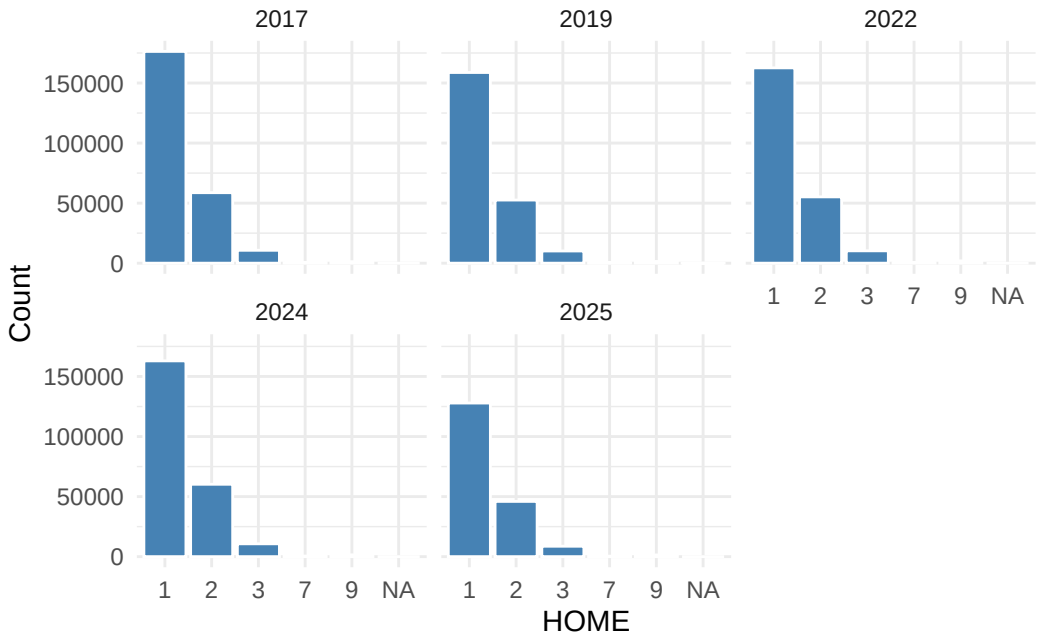
$RACE



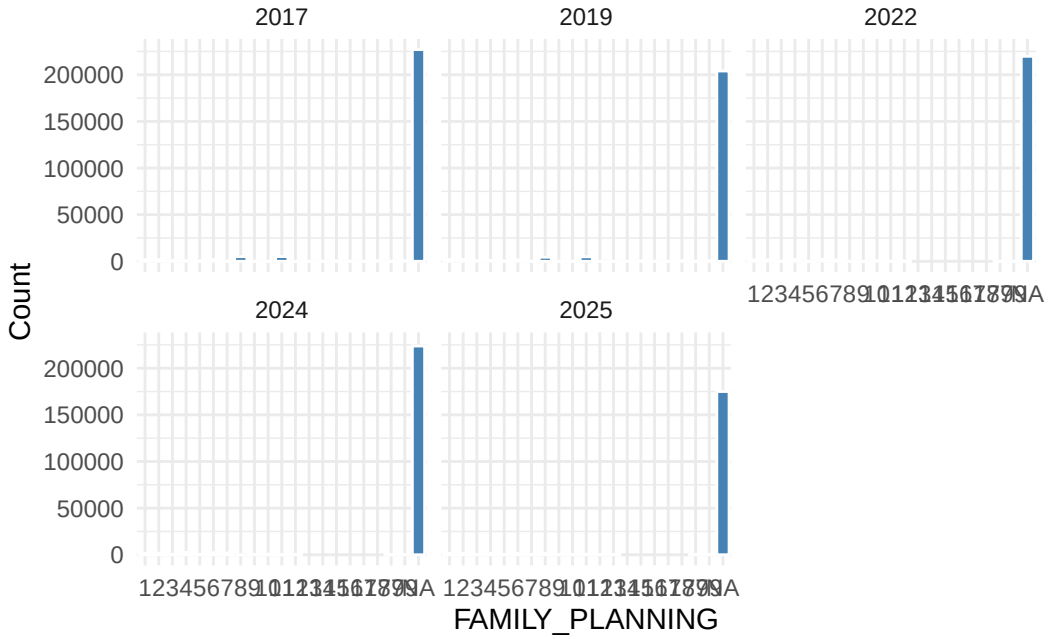
$MARITAL



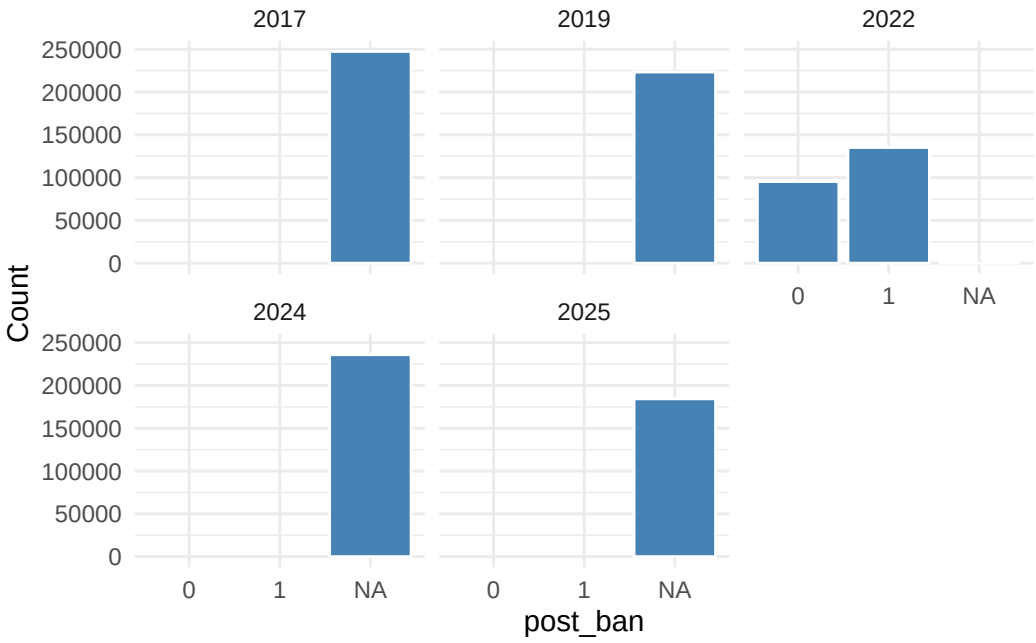
$HOME



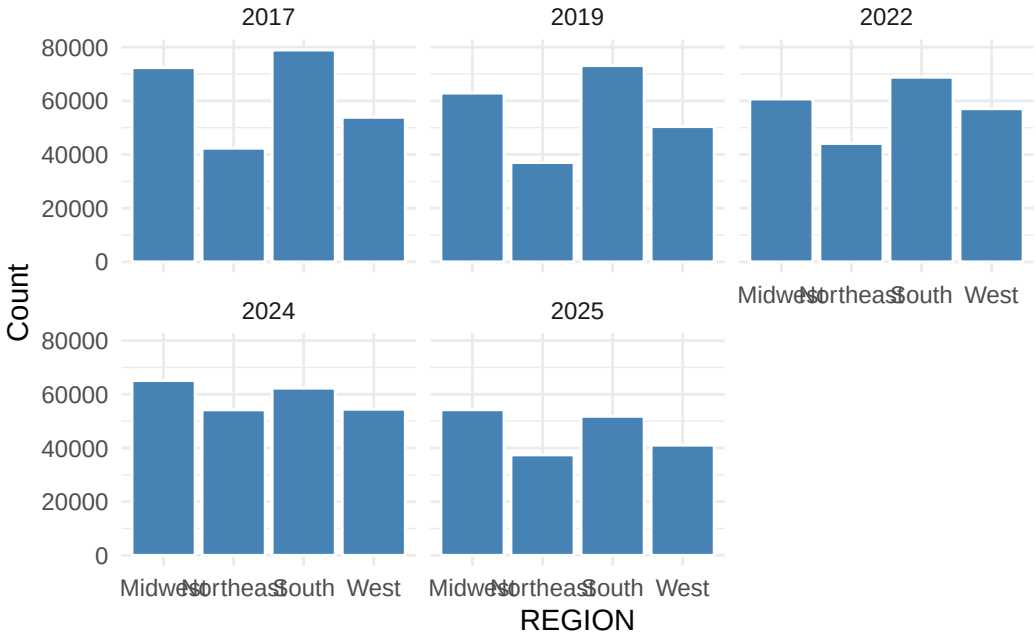
$FAMILY_PLANNING



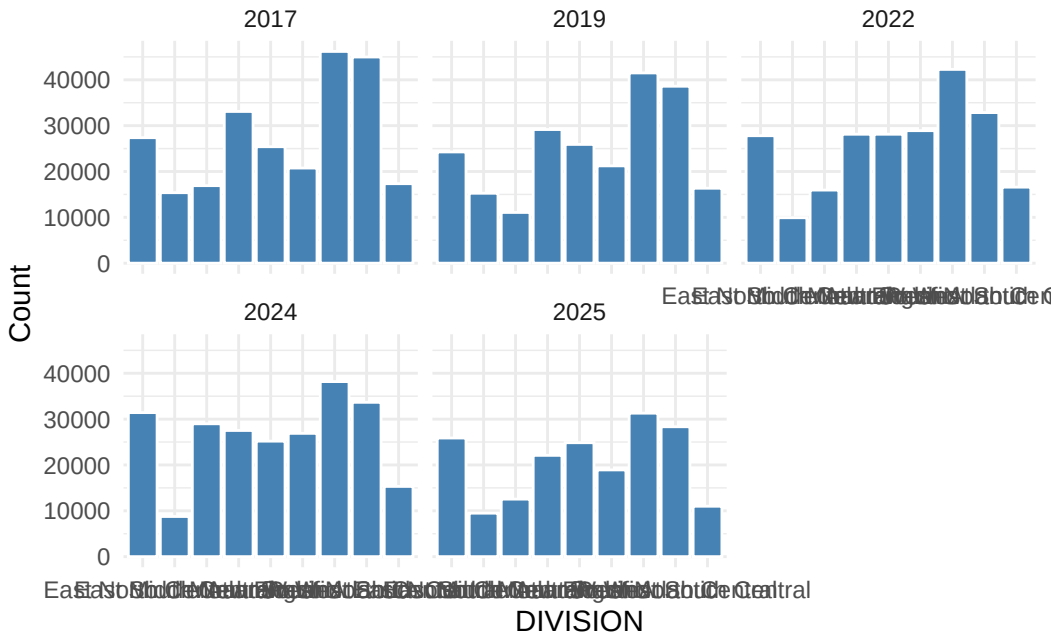
$post_ban



$REGION

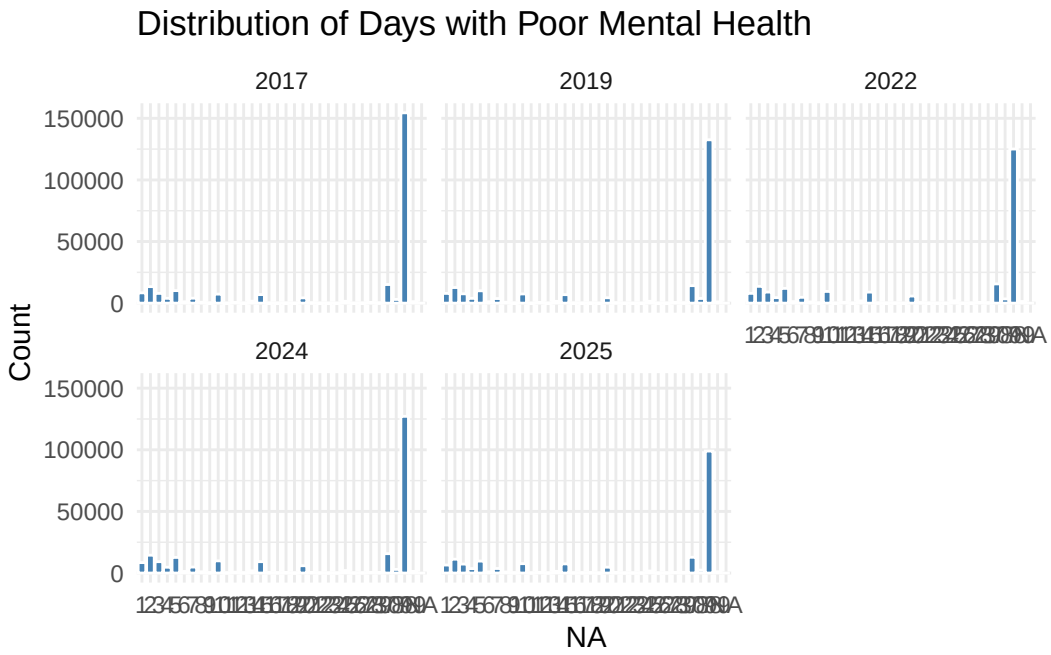


$DIVISION




```
# numerical variable
  ggplot(
  analysis_data_v2 ,
  aes(x = factor(MENTHLTH))
) +
  geom_bar(
    fill = "steelblue",
    color = "white"
  ) +
  facet_wrap(~ YEAR) +
  theme_minimal() +
  labs(
    title = "Distribution of Days with Poor Mental Health",
    x = var,
    y = "Count"
  )
```

```
Warning in label(labels[[nm]] %||% ""): NAs introduced by coercion
```




```
get_descriptive_stats_over_entire_period <- function(data, var) {

  tab <- data |>
    janitor::tabyl(
      {{ var }},
      show_na = TRUE
    )

  # The first column is the categorical variable
  category_col <- names(tab)[1]

  total_n <- sum(tab$n)

  valid_n <- sum(
    tab$n[!is.na(tab[[category_col]])]
  )

  tab |>
    dplyr::mutate(
      percent = 100 * n / total_n,
      valid_percent = dplyr::if_else(
        valid_n > 0 & !is.na(.data[[category_col]]),
        100 * n / valid_n,
        NA_real_
```

```
      )
    )
}
get_descriptive_stats_over_entire_period(analysis_data,PREGNANT)
```

```
 PREGNANT      n     percent valid_percent
        1  11432  1.00000175     3.0394636
        2 362558 31.71436619    96.3944922
        7    947  0.08283779     0.2517820
        9   1182  0.10339416     0.3142622
       NA 767079 67.09940010            NA
```

```
get_descriptive_stats_over_entire_period(analysis_data,MENTHLTH)
```

```
 MENTHLTH      n      percent valid_percent
        1  39555  3.460030546    3.46006384
        2  66084  5.780625928    5.78068155
        3  41539  3.633578785    3.63361375
        4  20903  1.828467160    1.82848475
        5  55130  4.822436708    4.82248311
        6   5731  0.501312983    0.50131781
        7  21584  1.888036893    1.88805506
        8   4351  0.380598986    0.38060265
        9    731  0.063943429    0.06394404
       10  42260  3.696647475    3.69668304
       11    300  0.026242173    0.02624243
       12   3036  0.265570794    0.26557335
       13    421  0.036826516    0.03682687
       14   8309  0.726820726    0.72682772
       15  39977  3.496944536    3.49697818
       16    697  0.060969316    0.06096990
       17    601  0.052571820    0.05257233
       18    785  0.068667020    0.06866768
       19    128  0.011196661    0.01119677
       20  25415  2.223149446    2.22317084
       21   1411  0.123425688    0.12342688
       22    400  0.034989564    0.03498990
       23    271  0.023705430    0.02370566
       24    259  0.022655743    0.02265596
       25   8695  0.760585655    0.76059297
```

```
      26    269  0.023530482    0.02353071
      27    593  0.051872029    0.05187253
      28   2362  0.206613378    0.20661537
      29   1140  0.099720258    0.09972122
      30  75139  6.572702192    6.57276544
      77  15937  1.394071718    1.39408513
      88 653219 57.139620608   57.14017042
      99   5955  0.520907139    0.52091215
      NA     11  0.000962213            NA
```

```
get_descriptive_stats_over_entire_period(analysis_data,EMPLOY)
```

```
 EMPLOY      n    percent valid_percent
      1 427657 37.4088303    37.6710651
      2  72200  6.3156164     6.3598887
      3  22839  1.9978167     2.0118213
      4  22202  1.9420958     1.9557098
      5  90549  7.9206752     7.9761989
      6  27679  2.4211904     2.4381629
      7 377824 33.0497429    33.2814207
      8  83087  7.2679448     7.3188929
      9  11203  0.9799702     0.9868398
     NA   7958  0.6961174            NA
```

```
get_descriptive_stats_over_entire_period(analysis_data,INCOME)
```

```
 INCOME      n   percent valid_percent
      1  77835  6.808532      6.808532
      2 125785 11.002906     11.002906
      3 108922  9.527833      9.527833
      4 126922 11.102364     11.102364
      5 329376 28.811807     28.811807
      6 100829  8.819907      8.819907
      7  31201  2.729273      2.729273
      9 242328 21.197378     21.197378
```

```
get_descriptive_stats_over_entire_period(analysis_data,EDUCA)
```

```
 EDUCA      n      percent valid_percent
```

```
     1  70463  6.163674184     6.1638144
     2 285253 24.952195508    24.9527630
     3 324998 28.428846097    28.4294927
     4 457453 40.015202966    40.0161131
     9   5005  0.437806924     0.4378169
    NA     26  0.002274322            NA
```

```
get_descriptive_stats_over_entire_period(analysis_data,INSURANCE)
```

```
 INSURANCE      n   percent valid_percent
         1 605179 52.937374     52.937374
         2  58134  5.085208      5.085208
         9 479885 41.977418     41.977418
```

The pooled pregnancy item includes respondents aged 50 and older who were not asked the question. The tables below restrict pregnancy comparisons to ages 18–49 and separate blank responses from explicit unknown or refused codes.

Trends in the U.S. Regions Over Time

```
regional_summary <- analysis_data_v2 |>
  select(
    STATE,
    REGION,
    YEAR,
    all_of(variables_to_summarize)
  ) |>
  pivot_longer(
    cols = all_of(variables_to_summarize),
    names_to = "VARIABLE",
    values_to = "VALUE"
  ) |>
  mutate(
    VALUE_CODE = if_else(
      is.na(VALUE),
      "Missing",
      as.character(VALUE)
    ),
```

```
    VALUE_LABEL = map2_chr(
      VARIABLE,
      VALUE_CODE,
      function(variable, code) {
        if (code == "Missing") {
          return("Missing")
        }

        label <- unname(variable_labels[[variable]][code])

        if (is.na(label)) {
          paste("Unlabeled code:", code)
        } else {
          label
        }
      }
    )
  ) |>
  count(
    STATE,
    REGION,
    YEAR,
    VARIABLE,
    VALUE_CODE,
    VALUE_LABEL,
    name = "N"
  ) |>
  group_by(STATE, REGION, YEAR, VARIABLE) |>
  mutate(
    PERCENT = round(100 * N / sum(N),2)
  ) |>
  ungroup()

regional_summary
```

```
# A tibble: 23,594 x 8
   STATE   REGION  YEAR VARIABLE VALUE_CODE VALUE_LABEL              N PERCENT
   <chr>   <chr>  <int> <chr>    <chr>      <chr>                <int>   <dbl>
 1 Alabama South   2017 AGEG5YR  1          18 to 24 years         182    4.51
 2 Alabama South   2017 AGEG5YR  10         65 to 69 years         471   11.7
 3 Alabama South   2017 AGEG5YR  11         70 to 74 years         429   10.6
```

```
 4 Alabama South   2017 AGEG5YR  12         75 to 79 years           310    7.68
 5 Alabama South   2017 AGEG5YR  13         80 years or older        356    8.82
 6 Alabama South   2017 AGEG5YR  14         Don't know/refused/mi~    53    1.31
 7 Alabama South   2017 AGEG5YR  2          25 to 29 years           160    3.96
 8 Alabama South   2017 AGEG5YR  3          30 to 34 years           226    5.6
 9 Alabama South   2017 AGEG5YR  4          35 to 39 years           225    5.57
10 Alabama South   2017 AGEG5YR  5          40 to 44 years           221    5.47
# i 23,584 more rows
```

```r
plot_variable <- function(data, variable_name, plot_title, legend_title) {

  data |>
    dplyr::filter(VARIABLE == variable_name) |>
    ggplot(
      aes(
        x = factor(YEAR),
        y = PERCENT,
        fill = VALUE_LABEL
      )
    ) +
    geom_col(position = "dodge") +
    facet_wrap(~ REGION) +
    labs(
      title = plot_title,
      x = "Year",
      y = "Percent",
      fill = legend_title
    ) +
    theme_minimal() +
    theme(
      legend.position = "bottom",
      axis.text.x = element_text(angle = 45, hjust = 1)
    )
}

var_by_region_plots <- list(
  marital = plot_variable(
    regional_summary,
    "MARITAL",
    "Marital Status by Region and Year",
    "Marital status"
  ),
```

```
home = plot_variable(
  regional_summary,
  "HOME",
  "Homeowner Status by Region and Year",
  "Homeowner status"
),

pregnant = plot_variable(
  regional_summary,
  "PREGNANT",
  "Pregnancy Status by Region and Year",
  "Pregnancy status"
),

employ = plot_variable(
  regional_summary,
  "EMPLOY",
  "Employment Status by Region and Year",
  "Employment status"
),

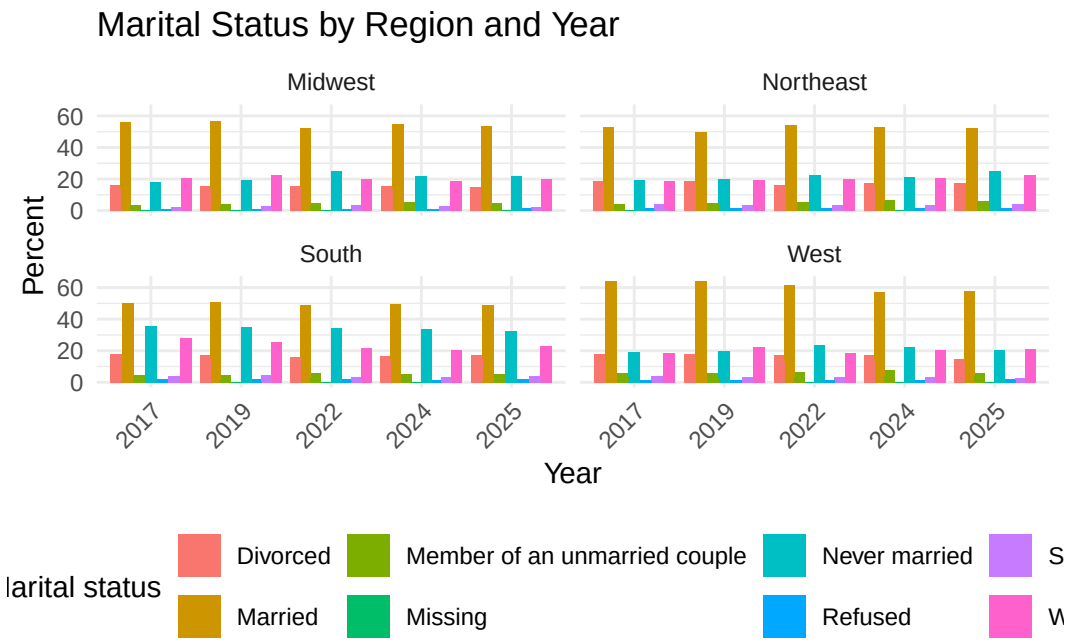
income = plot_variable(
  regional_summary,
  "INCOME",
  "Income Categories by Region and Year",
  "Income category"
),

insurance = plot_variable(
  regional_summary,
  "INSURANCE",
  "Health Insurance Status by Region and Year",
  "Insurance status"
),

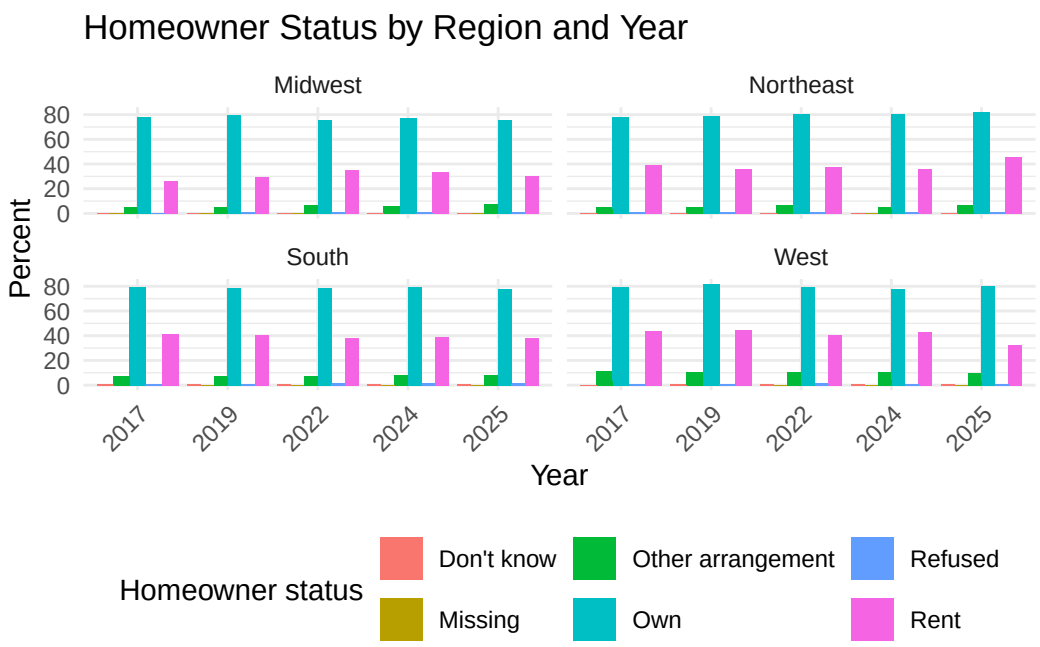
race = plot_variable(
  regional_summary,
  "RACE",
  "Race and Ethnicity by Region and Year",
  "Race and ethnicity"
)
```

```
)
```

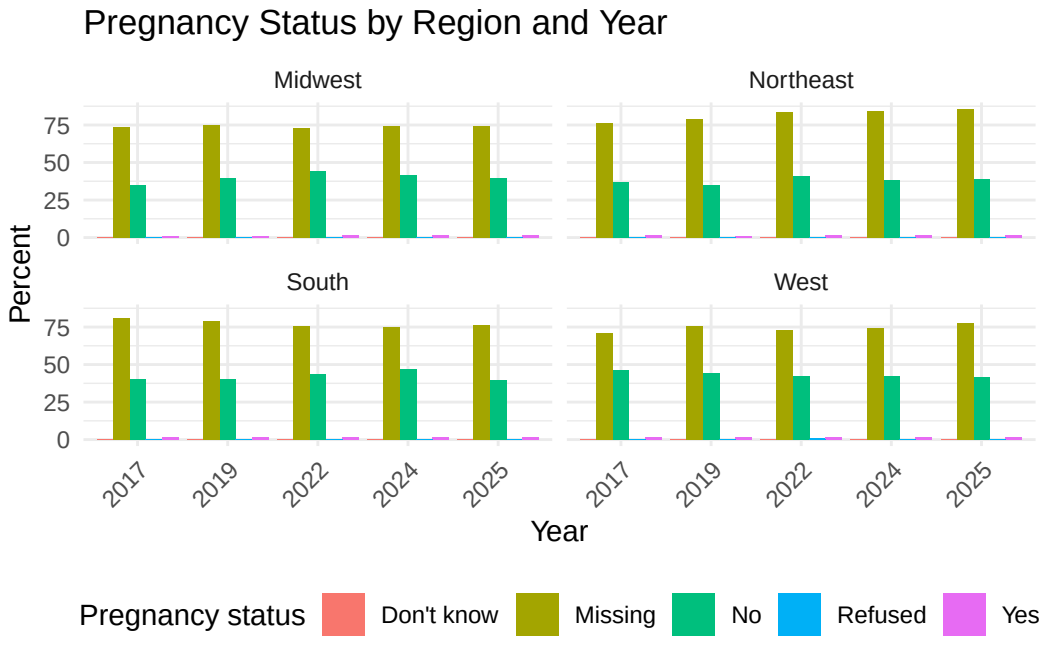
```
var_by_region_plots$marital
```



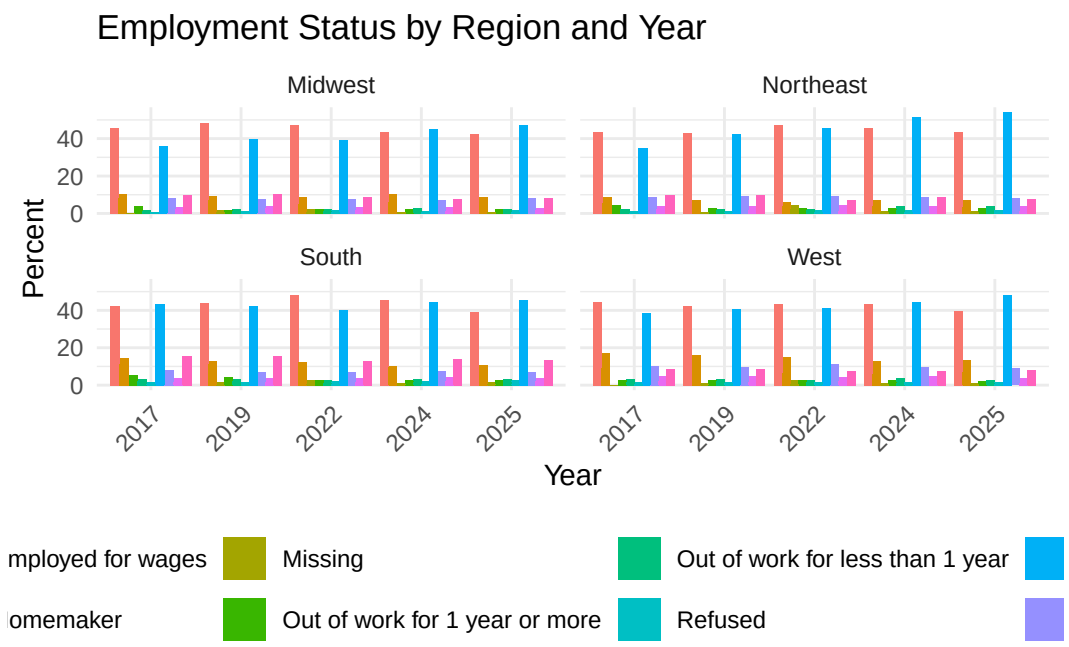

```
var_by_region_plots$home
```



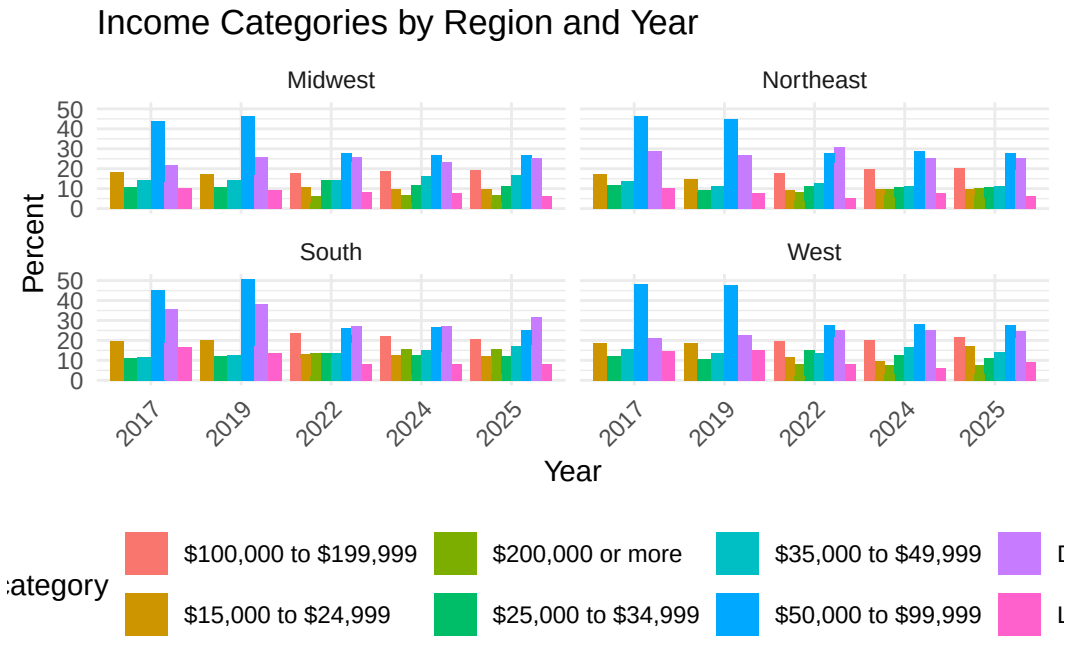

```
var_by_region_plots$pregnant
```



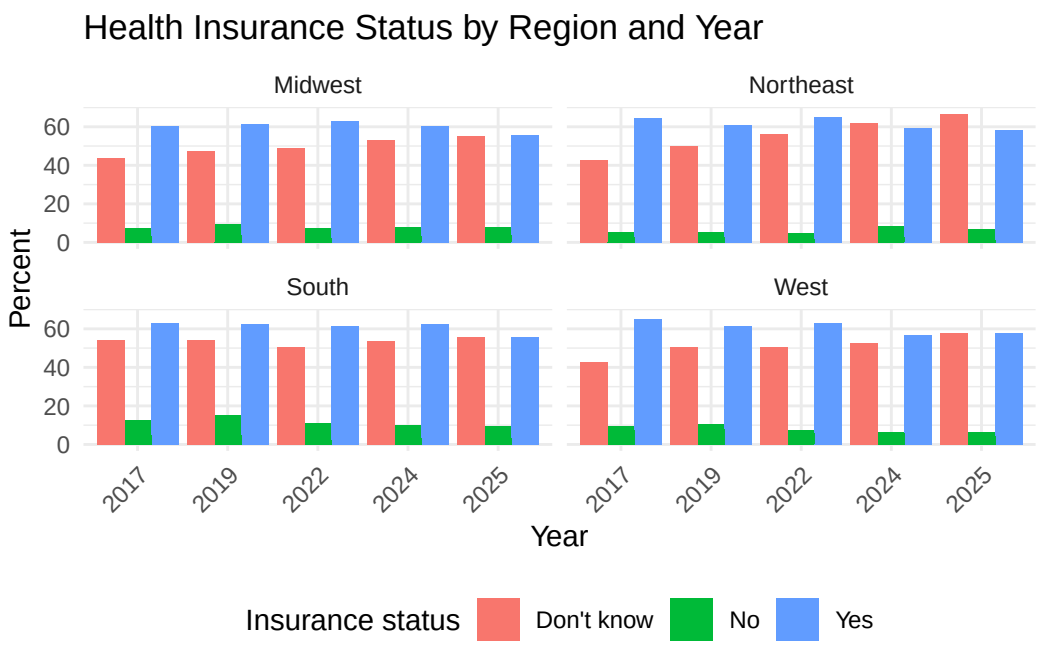

```
var_by_region_plots$employ
```



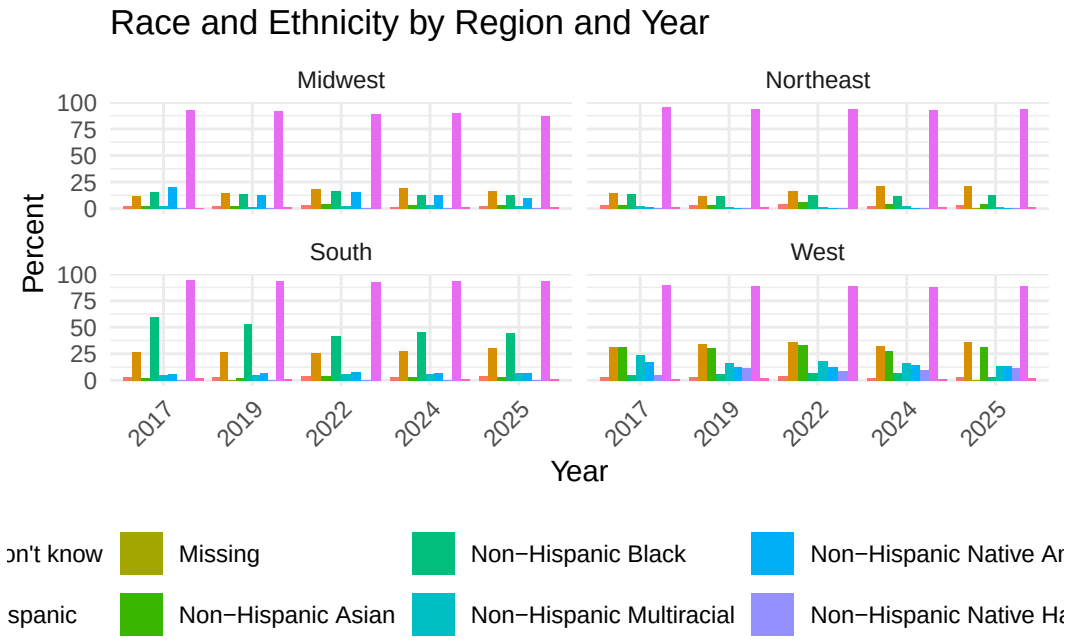

```
var_by_region_plots$income
```




```
var_by_region_plots$insurance
```




```
var_by_region_plots$race
```



NEW PLOTS:

Age distribution and pregnancy outcome by year (ages 18–49)

```
age_distribution_18_49 <- analysis_data_v2 |>
  filter(AGEG5YR %in% 1:6) |>
  count(AGEG5YR, name = "N") |>
  mutate(AGE_GROUP = unname(variable_labels$AGEG5YR[as.character(AGEG5YR)]),
         PERCENT = round(100 * N / sum(N), 1)) |>
  select(AGE_GROUP, N, PERCENT)

knitr::kable(age_distribution_18_49,
             caption = "Respondents in the analytic age range, pooled across
              ↪  included survey years")
```

Table 1: Respondents in the analytic age range, pooled across included survey years

AGE_GROUP	N	PERCENT
18 to 24 years	56190	15.2
25 to 29 years	49785	13.5
30 to 34 years	58841	15.9
35 to 39 years	65609	17.8
40 to 44 years	68688	18.6
45 to 49 years	70469	19.1

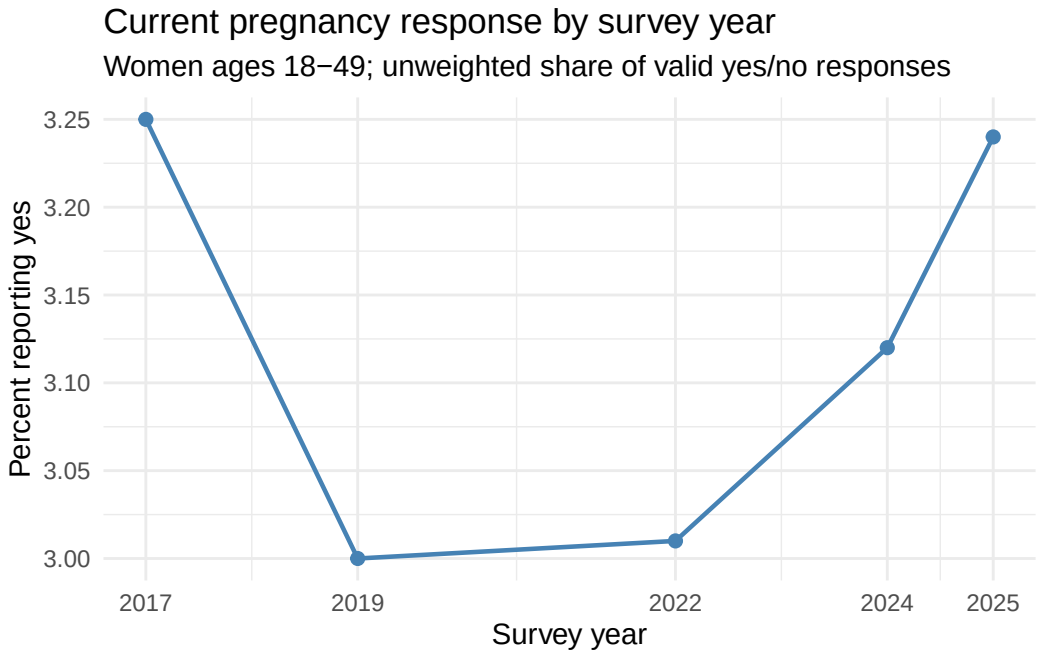
```
pregnancy_by_year_18_49 <- analysis_data_v2 |>
  filter(AGEG5YR %in% 1:6, PREGNANT %in% c(1, 2)) |>
  group_by(YEAR) |>
  summarise(
    eligible_responses = n(),
    reporting_current_pregnancy = sum(PREGNANT == 1),
    percent_reporting_pregnancy = round(100 * reporting_current_pregnancy /
     ↪  eligible_responses, 2),
    .groups = "drop"
  )

knitr::kable(pregnancy_by_year_18_49,
             caption = "Unweighted pregnancy responses among women ages
              ↪  18-49; descriptive only")
```

Table 2: Unweighted pregnancy responses among women ages 18–49; descriptive only

YEAR	eligible_responses	reporting_current_pregnancy	percent_reporting_pregnancy
2017	77153	2509	3.25
2019	68584	2056	3.00
2022	74586	2245	3.01
2024	77904	2429	3.12
2025	58563	1895	3.24

```
ggplot(pregnancy_by_year_18_49,
       aes(x = YEAR, y = percent_reporting_pregnancy)) +
  geom_line(color = "steelblue", linewidth = 0.8) +
  geom_point(color = "steelblue", size = 2) +
  scale_x_continuous(breaks = pregnancy_by_year_18_49$YEAR) +
  labs(title = "Current pregnancy response by survey year",
       subtitle = "Women ages 18-49; unweighted share of valid yes/no
        ↪  responses",
       x = "Survey year", y = "Percent reporting yes") +
  theme_minimal()
```



Nonresponse and special response codes

Blank values and explicit don't-know/refused codes are reported separately. Pregnancy comparisons are restricted to ages 18–49 because older respondents were not asked this item. These are unweighted, pooled descriptive rates; they are not population estimates.

```
nonresponse_codes <- list(
  PREGNANT = c(7, 9), MENTHLTH = c(77, 99), EMPLOY = c(9),
  INCOME = c(9), EDUCA = c(9), INSURANCE = c(9),
  RACE = c(9), MARITAL = c(9), HOME = c(7, 9)
)

age_eligible_data <- analysis_data_v2 |> filter(AGEG5YR %in% 1:6)

missingness_summary_18_49 <- purrr::imap_dfr(nonresponse_codes,
 ↪  function(codes, variable) {
  x <- age_eligible_data[[variable]]
  blank_n <- sum(is.na(x))
  special_n <- sum(!is.na(x) & x %in% codes)
  tibble(
    VARIABLE = variable,
    BLANK_N = blank_n,
    UNKNOWN_OR_REFUSED_N = special_n,
    TOTAL_UNREPORTED_N = blank_n + special_n,
    PERCENT_UNREPORTED = round(100 * (blank_n + special_n) / length(x), 1)
  )
}) |> arrange(desc(PERCENT_UNREPORTED))

knitr::kable(missingness_summary_18_49,
             caption = "Blank plus explicit unknown/refused codes, ages
              ↪  18-49")
```

Table 3: Blank plus explicit unknown/refused codes, ages 18–49

VARIABLE	BLANK_N	UNKNOWN_OR_REFUSED_N	TOTAL_UNREPORTED_N	PERCENT_UNREPORTED
INCOME	0	60198	60198	16.3
PREGNANT	11094	1698	12792	3.5

VARIABLE	BLANK_N	UNKNOWN_OR_REFUSED_N	TOTAL_UNREPORTED_N	PERCENT_UNREPORTED
INSURANCE	0	9672	9672	2.6
EMPLOY	3197	3929	7126	1.9
RACE	1	5790	5791	1.6
MENTHLTH	0	5403	5403	1.5
MARITAL	8	2286	2294	0.6
HOME	6	2359	2365	0.6
EDUCA	4	1111	1115	0.3

Family planning responses and missingness (2025)

The 2025 family-planning module is summarized separately because its questions and response categories do not line up exactly with the earlier survey waves. These summaries use women respondents in the 2025 file. Blank responses include structural skips and states that may not have included this module, so the tables distinguish full-sample blanks from blanks among age-eligible respondents in states with recorded module responses. The respondent eligibility follows questionnaire routing: ages 18–49 and not coded as currently pregnant, then the preceding answer where applicable. Counts and percentages below are unweighted.

```
family_planning_data <- sub_dat25 |>
  mutate(
    age_18_49 = `_AGEG5YR` %in% 1:6,
    not_coded_currently_pregnant = is.na(PREGNANT) | as.numeric(PREGNANT) !=
     ↪  1,
    module_eligible = age_18_49 & not_coded_currently_pregnant
  )

family_planning_labels <- list(
  HADSEX = c(
    `1` = "Yes", `2` = "No", `7` = "Don't know/not sure", `9` = "Refused"
  ),
  PREVENT_PREGNANCY = c(
    `1` = "Yes", `2` = "No", `7` = "Don't know/not sure", `9` = "Refused"
  ),
  FAMILY_PLANNING = c(
    `1` = "Female sterilization", `2` = "Male sterilization",
    `3` = "Implant", `4` = "IUD", `5` = "Shots",
    `6` = "Prescription pills, ring, or patch", `7` = "Over-the-counter
     ↪  pill",
```

```
    `8` = "Condoms", `9` = "Diaphragm, gel, cap, or similar",
    `10` = "Fertility awareness/rhythm", `11` = "Withdrawal",
    `12` = "Emergency contraception", `13` = "Other method",
    `77` = "Don't know/not sure", `99` = "Refused"
  ),
  NO_PREVENT_REASON = c(
    `1` = "No expected sex/no regular partner", `2` = "Did not think about
     ↪ it",
    `3` = "Wanted a pregnancy", `4` = "Did not care if pregnant",
    `5` = "Did not want to use birth control", `6` = "Trouble getting/paying
     ↪ for birth control",
    `8` = "Thought pregnancy was not possible", `9` = "Using withdrawal",
    `10` = "Female sterilization", `11` = "Partner vasectomy",
    `12` = "Breastfeeding/recent birth", `14` = "Other reason",
    `77` = "Don't know/not sure", `99` = "Refused"
  )
)

family_planning_distribution <- purrr::imap_dfr(family_planning_labels,
 ↪ function(labels, variable) {
  raw <- as.numeric(family_planning_data[[variable]])
  observed_codes <- ifelse(is.na(raw), "BLANK", as.character(raw))
  counts <- tibble(RESPONSE_CODE = observed_codes) |>
    count(RESPONSE_CODE, name = "N")
  response_levels <- c(names(labels), "BLANK")
  tibble(
    VARIABLE = variable,
    RESPONSE_CODE = response_levels,
    RESPONSE = c(unname(labels), "Not asked or blank")
  ) |>
    left_join(counts, by = "RESPONSE_CODE") |>
    mutate(
      N = tidyr::replace_na(N, 0L),
      PERCENT_OF_ALL_2025_WOMEN = round(100 * N /
       ↪ nrow(family_planning_data), 2),
      PERCENT_OF_NONBLANK_RESPONSES = if_else(
        RESPONSE_CODE == "BLANK", NA_real_,
        round(100 * N / sum(N[RESPONSE_CODE != "BLANK"]), 2)
      )
    )
})
```

```
knitr::kable(family_planning_distribution,
             caption = "Family-planning response counts and unweighted
              ↪  percentages, 2025 women respondents")
```

Table 4: Family-planning response counts and unweighted percentages, 2025 women respondents

VARIABLE	RESPONSE_CODE	RESPONSE	N	PERCENT_OF_ALL_2025_WOMEN	PERCENT_OF_NON-BLANK_RESPONSES
HADSEX	1	Yes	12191	6.50	72.03
HADSEX	2	No	3852	2.05	22.76
HADSEX	7	Don't know/not sure	40	0.02	0.24
HADSEX	9	Refused	842	0.45	4.97
HADSEX	BLANK	Not asked or blank	170771	90.98	NA
PREVENT_PREGNANCY	1	Yes	9432	5.03	77.71
PREVENT_PREGNANCY	2	No	2586	1.38	21.31
PREVENT_PREGNANCY	7	Don't know/not sure	30	0.02	0.25
PREVENT_PREGNANCY	9	Refused	89	0.05	0.73
PREVENT_PREGNANCY	BLANK	Not asked or blank	175559	93.53	NA
FAMILY_PLANNING	1	Female sterilization	1532	0.82	16.28
FAMILY_PLANNING	2	Male sterilization	1364	0.73	14.49
FAMILY_PLANNING	3	Implant	324	0.17	3.44

VARIABLE	RESPONSE_CODE	RESPONSE	N	PERCENT_OF_ALL_2025_WOMEN	PERCENT_OF_NON-BLANK_RESPONSES
FAMILY_PLANNING	4	IUD	1702	0.91	18.09
FAMILY_PLANNING	5	Shots	132	0.07	1.40
FAMILY_PLANNING	6	Prescription pills, ring, or patch	1319	0.70	14.02
FAMILY_PLANNING	7	Over-the-counter pill	181	0.10	1.92
FAMILY_PLANNING	8	Condoms	1411	0.75	14.99
FAMILY_PLANNING	9	Diaphragm, gel, cap, or similar	55	0.03	0.58
FAMILY_PLANNING	10	Fertility awareness/rhythm	176	0.09	1.87
FAMILY_PLANNING	11	Withdrawal	587	0.31	6.24
FAMILY_PLANNING	12	Emergency contraception	78	0.04	0.83
FAMILY_PLANNING	13	Other method	357	0.19	3.79
FAMILY_PLANNING	77	Don’t know/not sure	49	0.03	0.52
FAMILY_PLANNING	99	Refused	144	0.08	1.53

VARIABLE	RESPONSE_CODE	RESPONSE	N	PERCENT_OF_ALL_2025_WOMEN	PERCENT_OF_NON-BLANK_RESPONSES
FAMILY_PLANNING	BLANK	Not asked or blank	178285	94.99	NA
NO_PREVENT_REASON	1	No expected sex/no regular partner	63	0.03	2.45
NO_PREVENT_REASON	2	Did not think about it	123	0.07	4.79
NO_PREVENT_REASON	3	Wanted a pregnancy	617	0.33	24.04
NO_PREVENT_REASON	4	Did not care if pregnant	149	0.08	5.80
NO_PREVENT_REASON	5	Did not want to use birth control	30	0.02	1.17
NO_PREVENT_REASON	6	Trouble getting/paying for birth control	21	0.01	0.82
NO_PREVENT_REASON	8	Thought pregnancy was not possible	455	0.24	17.72
NO_PREVENT_REASON	9	Using withdrawal	25	0.01	0.97
NO_PREVENT_REASON	10	Female sterilization	297	0.16	11.57
NO_PREVENT_REASON	11	Partner vasectomy	71	0.04	2.77
NO_PREVENT_REASON	12	Breastfeeding/recent birth	28	0.01	1.09

VARIABLE	RESPONSE	RESPONSE	N	PERCENT_OF_ALL_2025_WOMEN	PERCENT_OF_NON-BLANK_RESPONSES
NO_PREVENT_REASON	14	Other reason	577	0.31	22.48
NO_PREVENT_REASON	77	Don't know/not sure	65	0.03	2.53
NO_PREVENT_REASON	99	Refused	46	0.02	1.79
NO_PREVENT_REASON	BLANK	Not asked or blank	185129	98.63	NA

```
state_module_coverage <- family_planning_data |>
  filter(module_eligible, !is.na(`_STATE`)) |>
  group_by(`_STATE`) |>
  summarise(
    AGE_ELIGIBLE_N = n(),
    HADSEX_RECORDED_N = sum(!is.na(HADSEX)),
    HADSEX_RECORDED_PERCENT = round(100 * HADSEX_RECORDED_N /
     ↪  AGE_ELIGIBLE_N, 1),
    .groups = "drop"
  ) |>
  arrange(HADSEX_RECORDED_PERCENT)

knitr::kable(state_module_coverage,
             caption = "State-level recorded response coverage for HADSEX
              ↪  among age/pregnancy-eligible women")
```

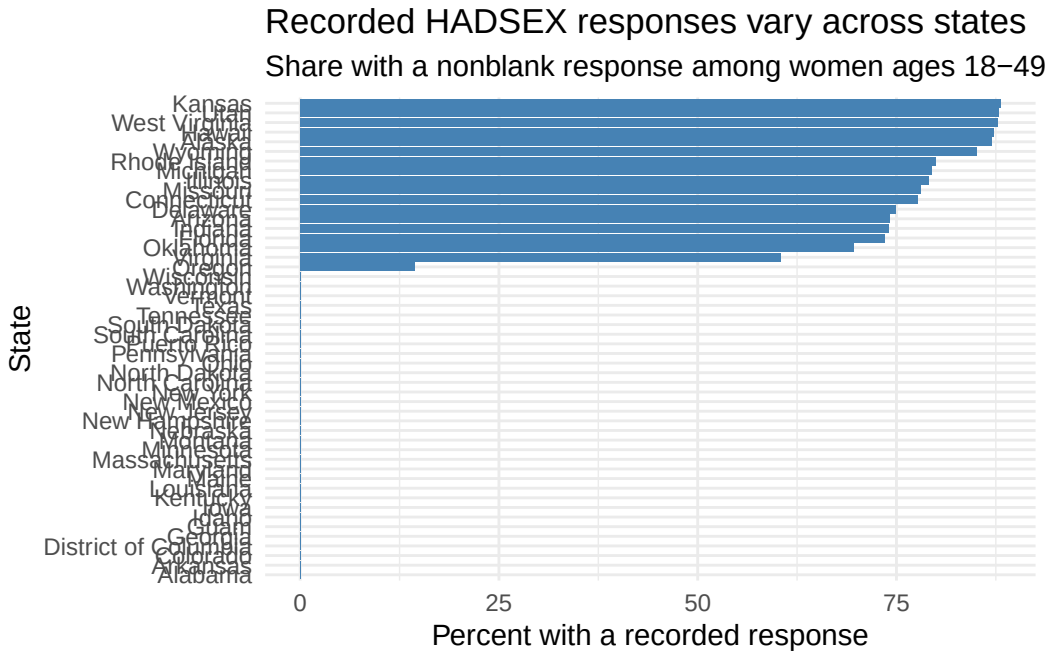
Table 5: State-level recorded response coverage for HADSEX among age/pregnancy-eligible women

_STATE	AGE_ELIGIBLE_N	HADSEX_RECORDED_N	HADSEX_RECORDED_PERCENT
Alabama	780	0	0.0

_STATE	AGE_ELIGIBLE_N	HAD-SEX_RECORDED_N	HAD-SEX_RECORDED_PER-CENT
Arkansas	762	0	0.0
Colorado	1856	0	0.0
District of Columbia	526	0	0.0
Georgia	1178	0	0.0
Guam	364	0	0.0
Idaho	655	0	0.0
Iowa	1395	0	0.0
Kentucky	1243	0	0.0
Louisiana	826	0	0.0
Maine	1570	0	0.0
Maryland	2046	0	0.0
Massachusetts	1685	0	0.0
Minnesota	2510	0	0.0
Montana	843	0	0.0
Nebraska	1324	0	0.0
New Hampshire	477	0	0.0
New Jersey	1372	0	0.0
New Mexico	626	0	0.0
New York	3102	0	0.0
North Carolina	735	0	0.0
North Dakota	826	0	0.0
Ohio	1310	0	0.0
Pennsylvania	554	0	0.0
Puerto Rico	979	0	0.0
South Carolina	604	0	0.0
South Dakota	546	0	0.0
Tennessee	1066	0	0.0
Texas	1222	0	0.0
Vermont	882	0	0.0
Washington	3365	0	0.0
Wisconsin	1491	0	0.0
Oregon	842	121	14.4
Virginia	719	434	60.4
Oklahoma	856	596	69.6
Florida	1734	1275	73.5
Indiana	1261	933	74.0
Arizona	678	503	74.2
Delaware	561	420	74.9
Connecticut	1437	1117	77.7

_STATE	AGE_ELIGIBLE_N	HADSEX_RECORDED_N	HADSEX_RECORDED_PERCENT
Missouri	1191	930	78.1
Illinois	2225	1759	79.1
Michigan	1492	1184	79.4
Rhode Island	876	700	79.9
Wyoming	503	428	85.1
Alaska	912	793	87.0
Hawaii	1228	1071	87.2
West Virginia	694	609	87.8
Utah	2341	2057	87.9
Kansas	1845	1626	88.1

```
ggplot(state_module_coverage,
       aes(x = reorder(`_STATE`, HADSEX_RECORDED_PERCENT), y =
        ↪  HADSEX_RECORDED_PERCENT)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(title = "Recorded HADSEX responses vary across states",
       subtitle = "Share with a nonblank response among women ages 18-49 not
        ↪  coded currently pregnant",
       x = "State", y = "Percent with a recorded response") +
  theme_minimal()
```



States with no recorded responses may not have included the optional module. Confirm state-level questionnaire coverage before interpreting those blanks as individual nonresponse.

```
states_with_family_planning_data <- state_module_coverage |>
  filter(HADSEX_RECORDED_N > 0) |>
  pull(`_STATE`)
family_planning_data$module_state_has_responses <-
 ↪  family_planning_data$`_STATE` %in% states_with_family_planning_data

question_eligibility <- list(
  HADSEX = family_planning_data$module_eligible &
   ↪  family_planning_data$module_state_has_responses,
  PREVENT_PREGNANCY = family_planning_data$module_eligible &
   ↪  family_planning_data$module_state_has_responses &
    !is.na(family_planning_data$HADSEX) &
     ↪  as.numeric(family_planning_data$HADSEX) == 1,
  FAMILY_PLANNING = family_planning_data$module_eligible &
   ↪  family_planning_data$module_state_has_responses &
    !is.na(family_planning_data$HADSEX) &
     ↪  as.numeric(family_planning_data$HADSEX) == 1 &
    !is.na(family_planning_data$PREVENT_PREGNANCY) &
     ↪  as.numeric(family_planning_data$PREVENT_PREGNANCY) == 1,
  NO_PREVENT_REASON = family_planning_data$module_eligible &
   ↪  family_planning_data$module_state_has_responses &
```

```
    !is.na(family_planning_data$HADSEX) &
     ↪  as.numeric(family_planning_data$HADSEX) == 1 &
    !is.na(family_planning_data$PREVENT_PREGNANCY) &
     ↪  as.numeric(family_planning_data$PREVENT_PREGNANCY) == 2
)

family_planning_missingness <- purrr::imap_dfr(question_eligibility,
 ↪  function(eligible, variable) {
  x <- as.numeric(family_planning_data[[variable]])
  special_codes <- if (variable %in% c("HADSEX", "PREVENT_PREGNANCY")) c(7,
 ↪  9) else c(77, 99)
  tibble(
    VARIABLE = variable,
    FULL_SAMPLE_N = length(x),
    FULL_SAMPLE_BLANK_N = sum(is.na(x)),
    FULL_SAMPLE_BLANK_PERCENT = round(100 * mean(is.na(x)), 1),
    ELIGIBLE_IN_STATES_WITH_DATA_N = sum(eligible, na.rm = TRUE),
    ELIGIBLE_BLANK_N = sum(eligible & is.na(x), na.rm = TRUE),
    ELIGIBLE_BLANK_PERCENT = round(100 * sum(eligible & is.na(x), na.rm =
     ↪  TRUE) / max(1, sum(eligible, na.rm = TRUE)), 1),
    ELIGIBLE_DK_OR_REFUSED_N = sum(eligible & !is.na(x) & x %in%
     ↪  special_codes, na.rm = TRUE)
  )
})

knitr::kable(family_planning_missingness,
             caption = "Full-sample blanks versus blanks among eligible
              ↪  respondents in states with recorded family-planning
              ↪  responses")
```

Table 6: Full-sample blanks versus blanks among eligible respondents in states with recorded family-planning responses

VARIABLE	FULL_SAMPLE_N	FULL_SAMPLE_BLANK_N	FULL_SAMPLE_BLANK_PERCENT	ELIGIBLE_IN_STATES_WITH_DATA_N	ELIGIBLE_BLANK_N	ELIGIBLE_BLANK_PERCENT	ELIGIBLE_DK_OR_REFUSED_N
HADSEX	187696	170771	91.0	21395	4839	22.6	764
PREVENT_PREGNANCY	187696	175559	93.5	12109	55	0.5	111

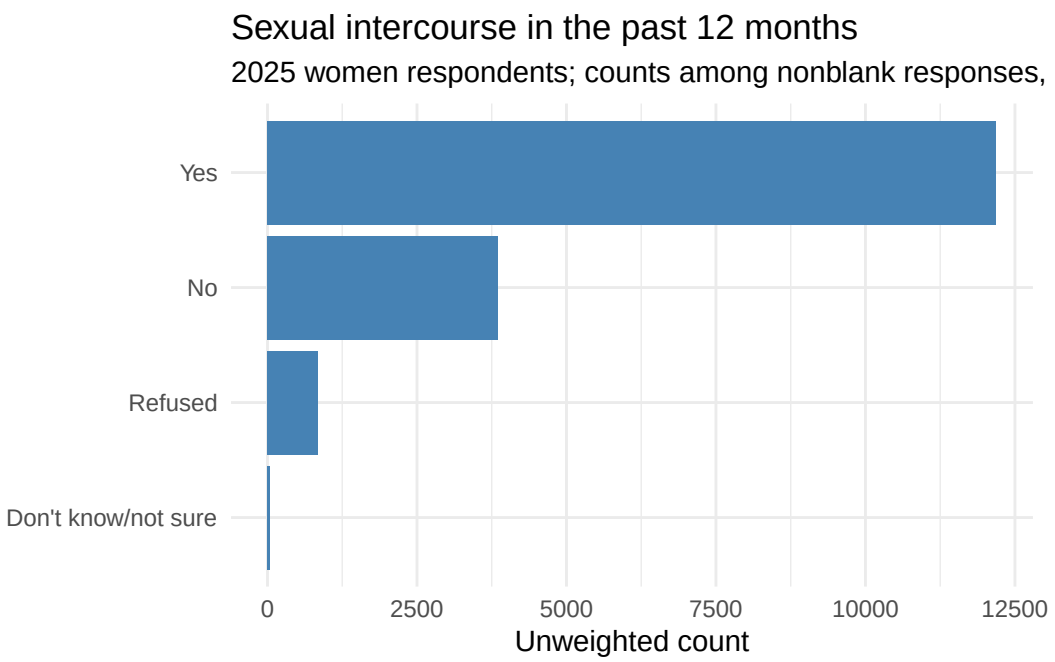
VARIABLE	FULL_SAMPLE_N	FULL_SAMPLE_BLANK_N	FULL_SAMPLE_BLANK_PERCENT	ELIGIBLE_IN_STATE	ELIGIBLE_BLANK_N	ELIGIBLE_BLANK_PERCENT	ELIGIBLE_DK_OR_REFUSED_N
FAMILY_PLANNING	187696	178285	95.0	9390	21	0.2	183
NO_PREVENT_REASON	187696	185129	98.6	2553	19	0.7	109

Bar charts for 2025 family-planning responses.

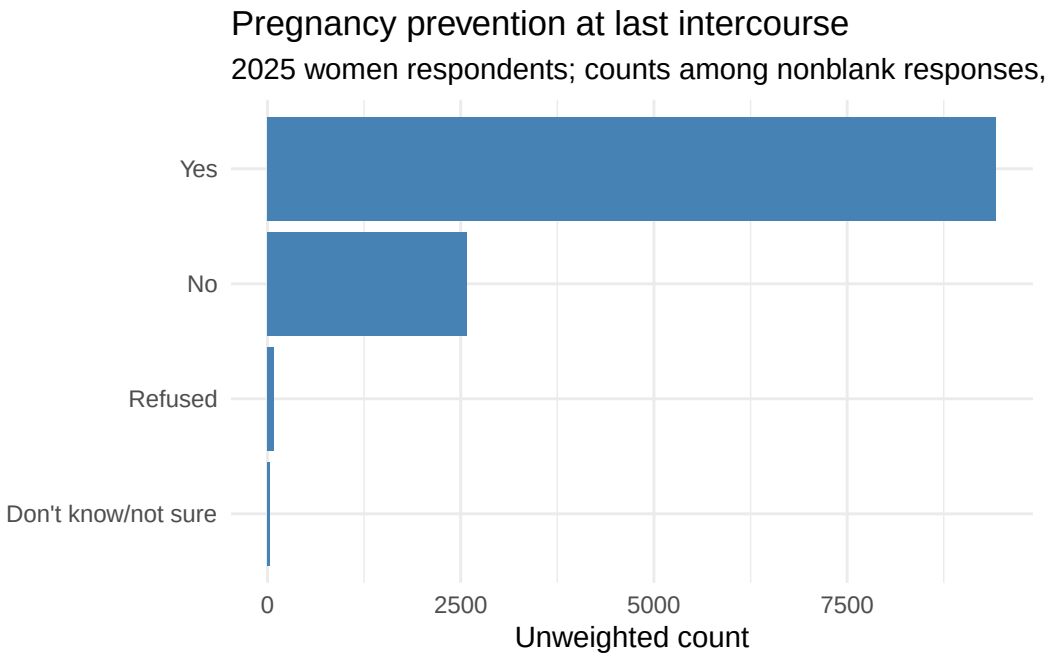
```
plot_family_planning_distribution <- function(variable, plot_title) {
  plot_data <- family_planning_distribution |>
    filter(VARIABLE == variable, RESPONSE_CODE != "BLANK") |>
    arrange(N) |>
    mutate(RESPONSE = factor(RESPONSE, levels = RESPONSE))

  ggplot(plot_data, aes(x = RESPONSE, y = N)) +
    geom_col(fill = "steelblue") +
    coord_flip() +
    labs(
      title = plot_title,
      subtitle = "2025 women respondents; counts among nonblank responses,
       ↪  including don't know/refused",
      x = NULL, y = "Unweighted count"
    ) +
    theme_minimal()
}

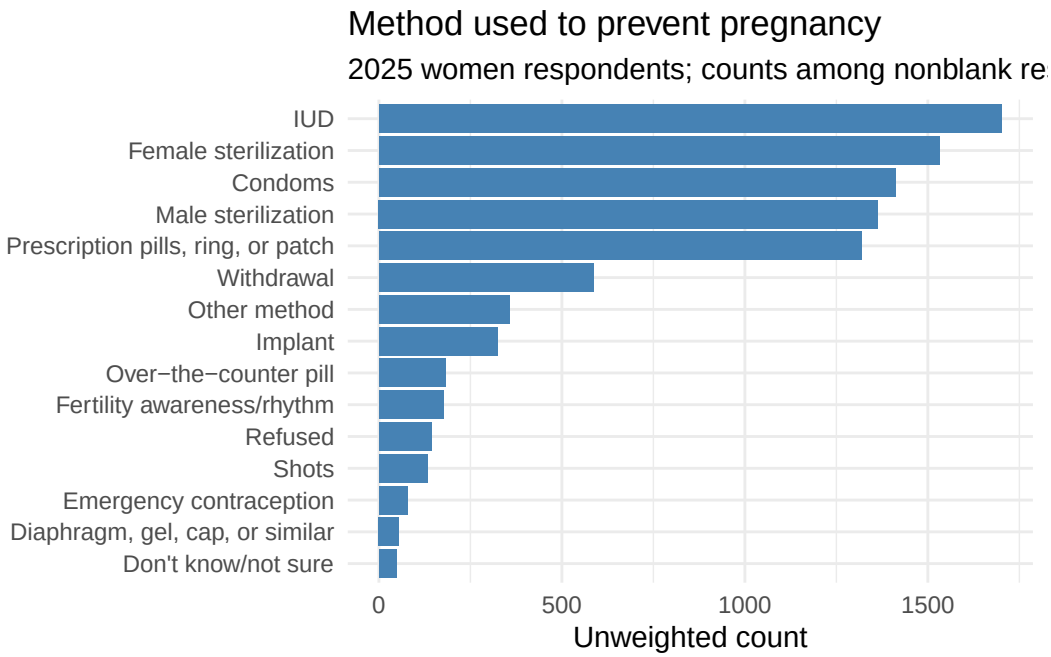
plot_family_planning_distribution("HADSEX", "Sexual intercourse in the past
 ↪  12 months")
```



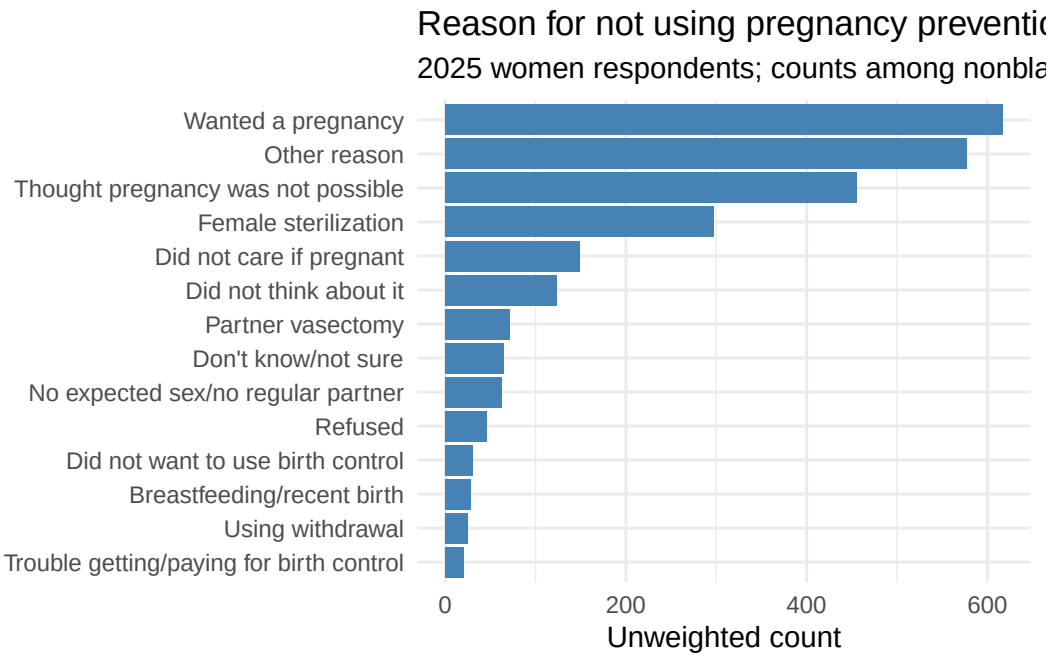

```
plot_family_planning_distribution("PREVENT_PREGNANCY", "Pregnancy prevention
 ↪  at last intercourse")
```



```
plot_family_planning_distribution("FAMILY_PLANNING", "Method used to prevent
 ↪  pregnancy")
```




```
plot_family_planning_distribution("NO_PREVENT_REASON", "Reason for not using
 ↪  pregnancy prevention")
```



EDA: DATA COMPLETENESS AND MISSINGNESS BY YEAR

```
# DATA COMPLETENESS AND MISSINGNESS BY YEAR

# Variables to assess
candidate_vars <- c(
  "PREGNANT",
  "MENTHLTH",
  "EMPLOY",
  "INCOME",
  "EDUCA",
  "INSURANCE",
  "RACE",
  "MARITAL",
  "HOME",
  "HADSEX","PFPPRVN4",
  "PREVENT_PREGNANCY",
  "FAMILY_PLANNING",
  "NOBCUSE9",
  "NO_PREVENT_REASON"
)
```

```r
available_vars <- intersect(candidate_vars, names(analysis_data))

# Harmonize aliases to avoid double-counting the same question
if ("PREVENT_PREGNANCY" %in% available_vars &&
    "PFPPRVN4" %in% available_vars) {
  available_vars <- setdiff(available_vars, "PREVENT_PREGNANCY")
}

if ("NO_PREVENT_REASON" %in% available_vars &&
    "NOBCUSE9" %in% available_vars) {
  available_vars <- setdiff(available_vars, "NO_PREVENT_REASON")
}

if (length(available_vars) == 0) {
  stop("None of the requested variables are available.")
}
```

Response Codes

```r
#  Special response codes: don't know/refused
special_codes <- list(
  PREGNANT = c(7, 9),
  MENTHLTH = c(77, 99),
  EMPLOY = c(9),
  INCOME = c(9),
  EDUCA = c(9),
  INSURANCE = c(9),
  RACE = c(9),
  MARITAL = c(9),
  HOME = c(7, 9),
  HADSEX = c(7, 9),
  PFPPRVN4 = c(7, 9),
  PREVENT_PREGNANCY = c(7, 9),
  FAMILY_PLANNING = c(77, 99),
  NOBCUSE9 = c(77, 99),
  NO_PREVENT_REASON = c(77, 99)
)
```

```r
# 3. Make a long dataset without mixing variable storage types
completeness_long <- analysis_data |>
  mutate(
    AGE_ELIGIBLE = `_AGEG5YR` %in% 1:6
  ) |>
  select(YEAR, AGE_ELIGIBLE, all_of(available_vars)) |>
  mutate(across(all_of(available_vars), as.character)) |>
  pivot_longer(
    cols = all_of(available_vars),
    names_to = "VARIABLE",
    values_to = "RESPONSE"
  ) |>
  mutate(
    RESPONSE = na_if(RESPONSE, ""),
    IS_BLANK = is.na(RESPONSE),
    IS_SPECIAL = map2_lgl(
      VARIABLE,
      RESPONSE,
      function(v, x) {
        codes <- special_codes[[v]]
        !is.na(x) && !is.null(codes) && x %in% as.character(codes)
      }
    )
  )

# 4. Overall completeness by year and variable
# Valid = nonblank and not explicitly don't know/refused.
# Blank values may include structural skips and unavailable modules.

completeness_by_year <- completeness_long |>
  group_by(YEAR, VARIABLE) |>
  summarise(
    TOTAL_N = n(),
    BLANK_N = sum(IS_BLANK),
    DK_REFUSED_N = sum(IS_SPECIAL),
    VALID_N = TOTAL_N - BLANK_N - DK_REFUSED_N,
    BLANK_PERCENT = round(100 * BLANK_N / TOTAL_N, 1),
    DK_REFUSED_PERCENT = round(100 * DK_REFUSED_N / TOTAL_N, 1),
    VALID_PERCENT = round(100 * VALID_N / TOTAL_N, 1),
    .groups = "drop"
  ) |>
```

```
  arrange(VARIABLE, YEAR)

knitr::kable(
  completeness_by_year,
  caption = paste(
    "Completeness and nonresponse by survey year;",
    "unweighted pooled records"
  )
)
```

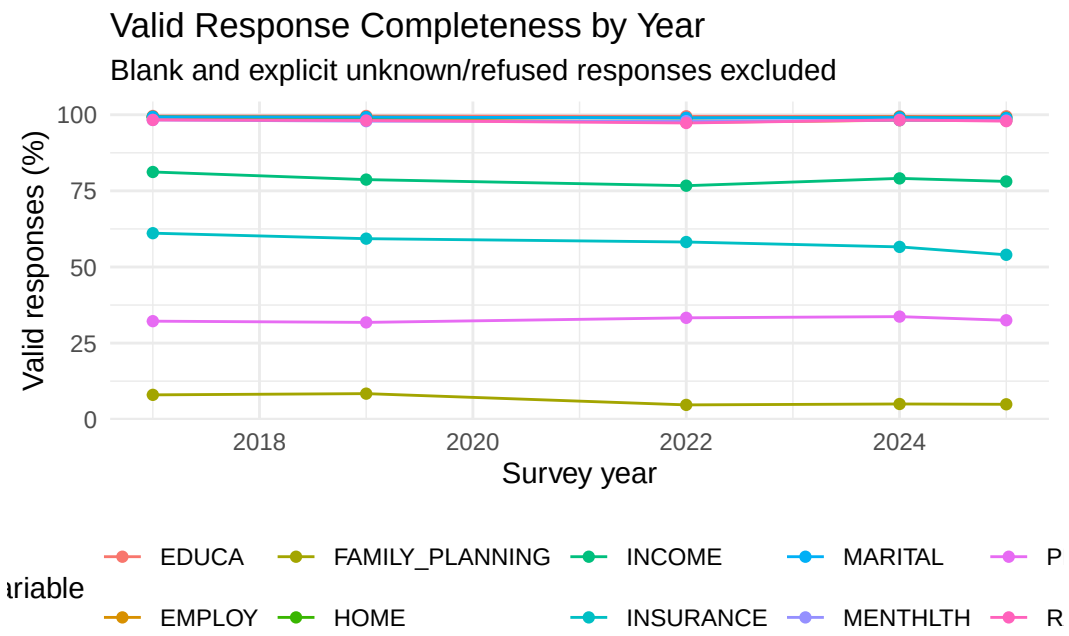
Table 7: Completeness and nonresponse by survey year; unweighted pooled records

YEAR	VARIABLE	TOTAL_N	BLANK_N	DK_REFUSED_N	VALID_N	BLANK_PERCENT	DK_REFUSED_PERCENT	VALID_PERCENT
2017	EDUCA	251007	4	916	250087	0.0	0.4	99.6
2019	EDUCA	228419	12	915	227492	0.0	0.4	99.6
2022	EDUCA	235893	4	1082	234807	0.0	0.5	99.5
2024	EDUCA	240183	4	1110	239069	0.0	0.5	99.5
2025	EDUCA	187696	2	982	186712	0.0	0.5	99.5
2017	EMPLOY	251007	4	1939	249064	0.0	0.8	99.2
2019	EMPLOY	228419	1556	1900	224963	0.7	0.8	98.5
2022	EMPLOY	235893	3228	2628	230037	1.4	1.1	97.5
2024	EMPLOY	240183	1747	2528	235908	0.7	1.1	98.2
2025	EMPLOY	187696	1423	2208	184065	0.8	1.2	98.1
2017	FAMILY_PLANNING	251007	230232	716	20059	91.7	0.3	8.0
2019	FAMILY_PLANNING	228419	208530	763	19126	91.3	0.3	8.4
2022	FAMILY_PLANNING	235893	224527	361	11005	95.2	0.2	4.7
2024	FAMILY_PLANNING	240183	228001	283	11899	94.9	0.1	5.0
2025	FAMILY_PLANNING	187696	178285	193	9218	95.0	0.1	4.9
2017	HOME	251007	3	1712	249292	0.0	0.7	99.3

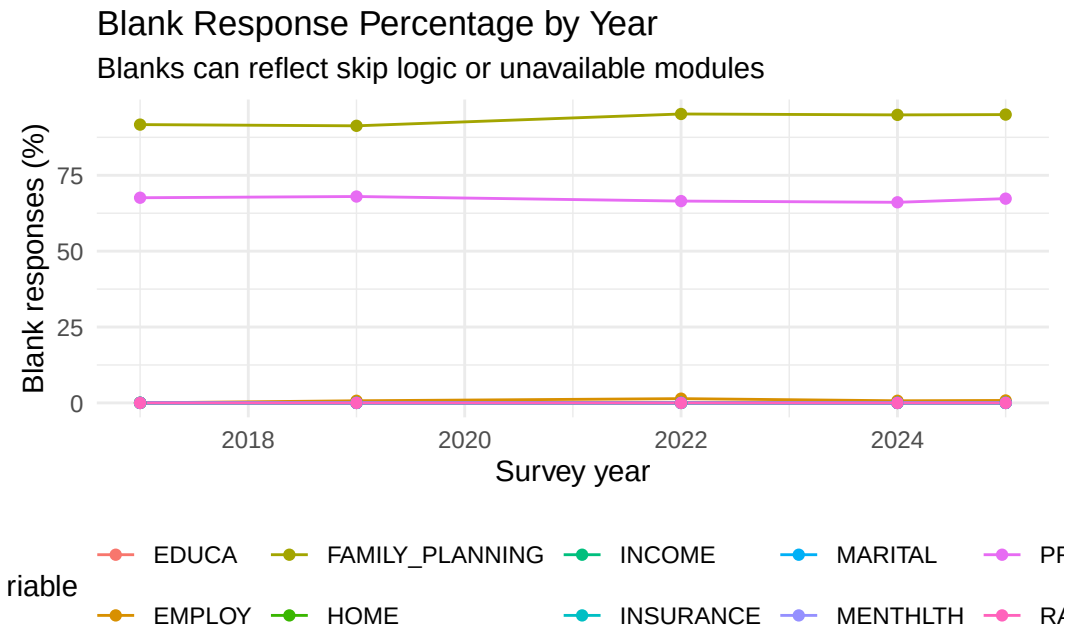
YEAR	VARIABLE	TOTAL_N	BLANK_N	DK_REFUSED_N	VALID_N	BLANK_PERCENT	DK_REFUSED_PERCENT	VALID_PERCENT
2019	HOME	228419	12	1871	226536	0.0	0.8	99.2
2022	HOME	235893	5	2473	233415	0.0	1.0	98.9
2024	HOME	240183	5	2015	238163	0.0	0.8	99.2
2025	HOME	187696	5	1848	185843	0.0	1.0	99.0
2017	INCOME	251007	0	47276	203731	0.0	18.8	81.2
2019	INCOME	228419	0	48551	179868	0.0	21.3	78.7
2022	INCOME	235893	0	55070	180823	0.0	23.3	76.7
2024	INCOME	240183	0	50261	189922	0.0	20.9	79.1
2025	INCOME	187696	0	41170	146526	0.0	21.9	78.1
2017	INSURANCE	251007	0	97757	153250	0.0	38.9	61.1
2019	INSURANCE	228419	0	92959	135460	0.0	40.7	59.3
2022	INSURANCE	235893	0	98576	137317	0.0	41.8	58.2
2024	INSURANCE	240183	0	104307	135876	0.0	43.4	56.6
2025	INSURANCE	187696	0	86286	101410	0.0	46.0	54.0
2017	MARITAL	251007	4	1759	249244	0.0	0.7	99.3
2019	MARITAL	228419	25	1840	226554	0.0	0.8	99.2
2022	MARITAL	235893	5	2432	233456	0.0	1.0	99.0
2024	MARITAL	240183	6	2089	238088	0.0	0.9	99.1
2025	MARITAL	187696	2	2024	185670	0.0	1.1	98.9
2017	MENTHLTH	251007	1	4292	246714	0.0	1.7	98.3
2019	MENTHLTH	228419	5	4904	223510	0.0	2.1	97.9
2022	MENTHLTH	235893	3	4880	231010	0.0	2.1	97.9
2024	MENTHLTH	240183	1	4324	235858	0.0	1.8	98.2
2025	MENTHLTH	187696	1	3492	184203	0.0	1.9	98.1
2017	PREGNANT	251007	169695	386	80926	67.6	0.2	32.2
2019	PREGNANT	228419	155337	379	72703	68.0	0.2	31.8
2022	PREGNANT	235893	156905	538	78450	66.5	0.2	33.3
2024	PREGNANT	240183	158842	463	80878	66.1	0.2	33.7
2025	PREGNANT	187696	126300	363	61033	67.3	0.2	32.5

YEAR	VARIABLE	TOTAL_N	BLANK_N	DK_REFUSED_N	VALID_N	BLANK_PERCENT	DK_REFUSED_PERCENT	VALID_PERCENT
2017	RACE	251007	0	4270	246737	0.0	1.7	98.3
2019	RACE	228419	3	4363	224053	0.0	1.9	98.1
2022	RACE	235893	0	6278	229615	0.0	2.7	97.3
2024	RACE	240183	0	4083	236100	0.0	1.7	98.3
2025	RACE	187696	2	3848	183846	0.0	2.1	97.9

```
# 5. Graph: valid response percentage by year
ggplot(
  completeness_by_year,
  aes(
    x = YEAR,
    y = VALID_PERCENT,
    color = VARIABLE,
    group = VARIABLE
  )
) +
  geom_line() +
  geom_point() +
  labs(
    title = "Valid Response Completeness by Year",
    subtitle = "Blank and explicit unknown/refused responses excluded",
    x = "Survey year",
    y = "Valid responses (%)",
    color = "Variable"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")
```

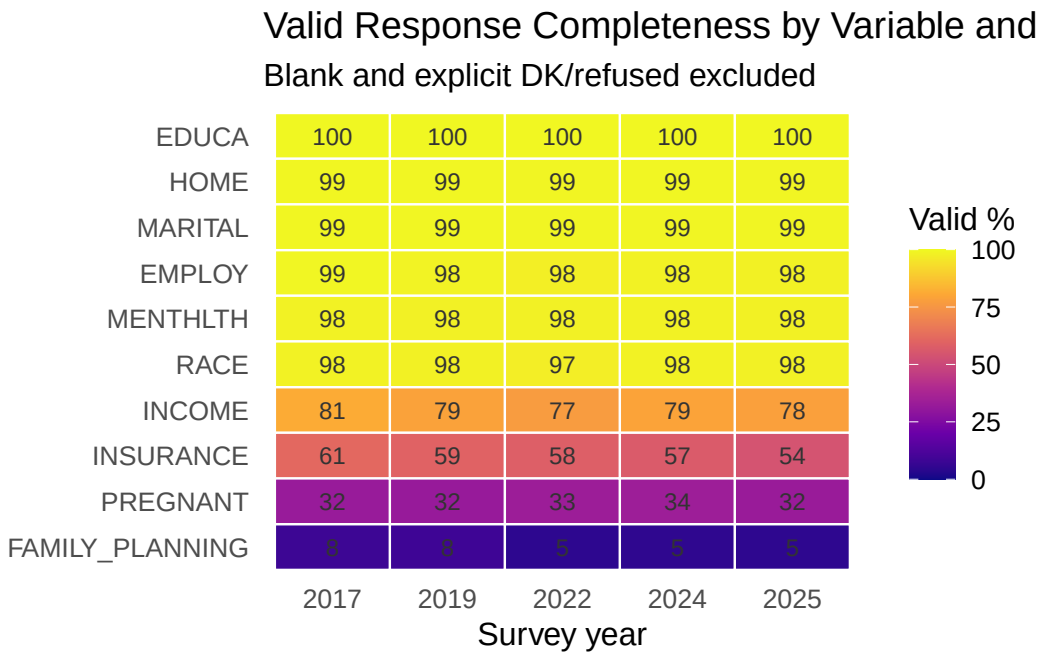



```
# 6. Graph: blank percentage by year
ggplot(
  completeness_by_year,
  aes(
    x = YEAR,
    y = BLANK_PERCENT,
    color = VARIABLE,
    group = VARIABLE
  )
) +
  geom_line() +
  geom_point() +
  labs(
    title = "Blank Response Percentage by Year",
    subtitle = "Blanks can reflect skip logic or unavailable modules",
    x = "Survey year",
    y = "Blank responses (%)",
    color = "Variable"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")
```



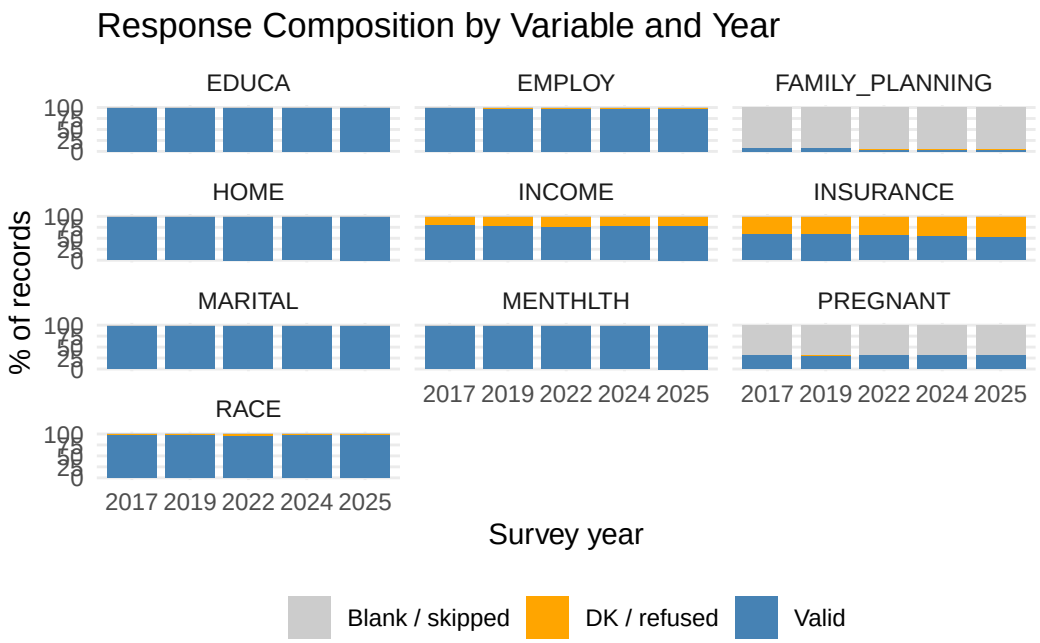
HeatMap for overall population

```
# Valid % heatmap
ggplot(completeness_by_year,
       aes(x = factor(YEAR), y = fct_reorder(VARIABLE, VALID_PERCENT, .fun =
        ↪  mean),
           fill = VALID_PERCENT)) +
  geom_tile(color = "white", linewidth = 0.4) +
  geom_text(aes(label = round(VALID_PERCENT)), size = 3, color = "grey20") +
  scale_fill_viridis_c(option = "C", limits = c(0, 100), na.value =
   ↪  "grey90") +
  labs(title = "Valid Response Completeness by Variable and Year",
       subtitle = "Blank and explicit DK/refused excluded",
       x = "Survey year", y = NULL, fill = "Valid %") +
  theme_minimal(base_size = 12) +
  theme(panel.grid = element_blank())
```



Stacked bar chart of response composition by variable and year

```
completeness_by_year |>
  select(YEAR, VARIABLE, VALID_PERCENT, DK_REFUSED_PERCENT, BLANK_PERCENT) |>
  pivot_longer(ends_with("PERCENT"), names_to = "STATUS", values_to = "PCT")
   ↪  |>
  mutate(STATUS = factor(STATUS,
                         levels = c("BLANK_PERCENT","DK_REFUSED_PERCENT","VA
                          ↪  LID_PERCENT"),
                         labels = c("Blank / skipped","DK /
                          ↪  refused","Valid"))) |>
  ggplot(aes(factor(YEAR), PCT, fill = STATUS)) +
  geom_col(width = 0.85) +
  facet_wrap(~ VARIABLE, ncol = 3) +
  scale_fill_manual(values = c("Blank / skipped" = "grey80",
                               "DK / refused" = "orange",
                               "Valid" = "steelblue")) +
  labs(title = "Response Composition by Variable and Year",
       x = "Survey year", y = "% of records", fill = NULL) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "bottom", panel.grid.minor = element_blank())
```




```
composition_table <- completeness_by_year |>
  select(YEAR, VARIABLE, VALID_PERCENT, DK_REFUSED_PERCENT, BLANK_PERCENT) |>
  rename(
    `Valid %`      = VALID_PERCENT,
    `DK/Refused %` = DK_REFUSED_PERCENT,
    `Blank/Skip %` = BLANK_PERCENT
  ) |>
  arrange(VARIABLE, YEAR)

knitr::kable(
  composition_table,
  caption = "Response composition (%) by variable and survey year",
  digits  = 1
)
```

Table 8: Response composition (%) by variable and survey year

YEAR	VARIABLE	Valid %	DK/Refused %	Blank/Skip %
2017	EDUCA	99.6	0.4	0.0
2019	EDUCA	99.6	0.4	0.0
2022	EDUCA	99.5	0.5	0.0
2024	EDUCA	99.5	0.5	0.0
2025	EDUCA	99.5	0.5	0.0

YEAR	VARIABLE	Valid %	DK/Refused %	Blank/Skip %
2017	EMPLOY	99.2	0.8	0.0
2019	EMPLOY	98.5	0.8	0.7
2022	EMPLOY	97.5	1.1	1.4
2024	EMPLOY	98.2	1.1	0.7
2025	EMPLOY	98.1	1.2	0.8
2017	FAMILY_PLANNING	8.0	0.3	91.7
2019	FAMILY_PLANNING	8.4	0.3	91.3
2022	FAMILY_PLANNING	4.7	0.2	95.2
2024	FAMILY_PLANNING	5.0	0.1	94.9
2025	FAMILY_PLANNING	4.9	0.1	95.0
2017	HOME	99.3	0.7	0.0
2019	HOME	99.2	0.8	0.0
2022	HOME	98.9	1.0	0.0
2024	HOME	99.2	0.8	0.0
2025	HOME	99.0	1.0	0.0
2017	INCOME	81.2	18.8	0.0
2019	INCOME	78.7	21.3	0.0
2022	INCOME	76.7	23.3	0.0
2024	INCOME	79.1	20.9	0.0
2025	INCOME	78.1	21.9	0.0
2017	INSURANCE	61.1	38.9	0.0
2019	INSURANCE	59.3	40.7	0.0
2022	INSURANCE	58.2	41.8	0.0
2024	INSURANCE	56.6	43.4	0.0
2025	INSURANCE	54.0	46.0	0.0
2017	MARITAL	99.3	0.7	0.0
2019	MARITAL	99.2	0.8	0.0
2022	MARITAL	99.0	1.0	0.0
2024	MARITAL	99.1	0.9	0.0
2025	MARITAL	98.9	1.1	0.0
2017	MENTHLTH	98.3	1.7	0.0
2019	MENTHLTH	97.9	2.1	0.0
2022	MENTHLTH	97.9	2.1	0.0
2024	MENTHLTH	98.2	1.8	0.0
2025	MENTHLTH	98.1	1.9	0.0
2017	PREGNANT	32.2	0.2	67.6
2019	PREGNANT	31.8	0.2	68.0
2022	PREGNANT	33.3	0.2	66.5
2024	PREGNANT	33.7	0.2	66.1
2025	PREGNANT	32.5	0.2	67.3
2017	RACE	98.3	1.7	0.0

YEAR	VARIABLE	Valid %	DK/Refused %	Blank/Skip %
2019	RACE	98.1	1.9	0.0
2022	RACE	97.3	2.7	0.0
2024	RACE	98.3	1.7	0.0
2025	RACE	97.9	2.1	0.0

SANITY CHECK: AGE AND PREGNANCY total numbers

```
analysis_data <- analysis_data |>
  mutate(
    # _AGEG5YR: 1=18-24, 2=25-29, 3=30-34, 4=35-39, 5=40-44, 6=45-49,
    #           7=50-54, 8=55-59, ...
    AGE_18_49    = `_AGEG5YR` %in% 1:6,
    AGE_18_59    = `_AGEG5YR` %in% 1:8,
    NOT_PREGNANT = is.na(PREGNANT) | PREGNANT != 1
  )

# Now the sanity check works
analysis_data |>
  summarise(
    total_rows         = n(),
    women              = n(),
    age_18_49          = sum(AGE_18_49),
    age_18_49_not_preg = sum(AGE_18_49 & NOT_PREGNANT),
    preg_yes           = sum(PREGNANT == 1, na.rm = TRUE),
    preg_dk_refused    = sum(PREGNANT %in% c(7, 9), na.rm = TRUE),
    preg_blank         = sum(is.na(PREGNANT))
  )
```

```
# A tibble: 1 x 7
  total_rows   women age_18_49 age_18_49_not_preg preg_yes preg_dk_refused
       <int>   <int>     <int>              <int>    <int>           <int>
1    1143198 1143198    378431             367064    11432            2129
# i 1 more variable: preg_blank <int>
```